An integrated resource planning model for utilities with quantified outage costs

by

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1 INTRODUCTION

1.1 Introduction

The utility industry has changed rapidly in the 1990's. For utilities to survive in a deregulated, competitive environment, to find efficient and effective ways to meet ever-increasing customer demand is crucial for utilities. A critical component of meeting customer demand is via integrated resource planning, and there have been numerous studies (e.g., Ref., Hobbs [1995a]).

However, the studies in the literature seldom consider some hard-to-quantify costs such as outage costs in their models. This will distort the accuracy of estimating the total cost to a utility. This provides us an important motivation to develop a more accurate model with easy-to-quantify as well as hard-to-quantify factors. In this thesis, we explicitly study how to quantify the cost of outage from customers' perspectives and utilities' perspectives. In short, our research is to reflect the real cost of a utility to make the capital budgeting process more accurate.

Because our research is based on the concepts of past system plannings, there are many fields of the utility industry that have been related to our research. In this chapter, we provide a comprehensive overview for the related areas of the utility industry as the background of this thesis. Subsequently, the organization of this thesis is described.

1.2 The Waves of Change in Electric Industry

In recent years, the policies of deregulation have dramatically changed the environment of the electric industry. Before the implementation of deregulation, utilities have the duty to serve the customers who live in their geographic area. On the other hand, the customers

should purchase power from the utilities located in their geographic areas. Under this regulated environment, utilities and customers did not have many choices to do anything really beneficial for themselves. However, the waves of change for electric utilities are coming after the arrival of the era of deregulation. Under these new rules, utilities do not have an obligation to serve the customers living in their geographical areas, if the business is not beneficial for utilities. On the other hand, based on the prices and quality of electric power service, customers might decide to purchase power from the utilities outside their geographic areas. For this dramatic change, we prefer to use the word reregulation instead of deregulation because now the rules of the game are regulated by the market itself.

The perspectives for the future of customers are that they will be in the driver's seat and can choose the sources of power. The impetus brought by competition will provide customers with lower costs. For utilities, they will face a really competitive environment and only those who know how to decrease their costs while satisfying customer demand can survive in such a cruel environment.

1.3 Nature of Generation Expansion Planning

Traditionally, to incorporate the rapidly increasing demand of customers, utilities have invested efforts to find more effective and efficient ways to plan electric power generation expansion. Because the budget needed for generation expansion planning is enormous, a large number of mathematical models have been developed for achieving the goals of minimizing the total cost of a utility.

In reality, there are lots of investment options for a utility to achieve its generation expansion planning. The main sources are hydro schemes and fossil fuels including oil, gas, fuel oil, thermal and gas turbines. Nuclear plants had been one of the main power sources. Because of the environmental concerns, nuclear plants may not be the possible power sources for utilities in the near future unless nuclear technology is dramatically improved and can be trusted by customers. However, different types of power plants have different cost structures or features. Anderson [1972] states, "gas turbines have low capital but high generation costs; fossil, higher

capital but lower generation costs; nuclear, high capital and low generation costs; and hydro, high or low capital costs and near-zero generation costs." Hence, utilities should determine an optimal combination for constructing new generation units while achieving the goals of least cost investments and satisfying many restrictions such as reliability concerns.

Briefly, generation expansion planning is a process of determining the optimal expansion strategies over a given planning horizon that may span fifteen to thirty years. More specifically, generation expansion planning is a problem of determining when facilities should be provided, what type of generation units should be constructed, and where the new generation units should be located. All these optimal processes are to provide electric power service with good quality and quantity.

Since numerous issues need to be considered in generation expansion planning over a long period of time, generation expansion planning becomes a challenging problem due to the complexities. Moreover, the difficulties of solving this type of problem may be raised because data and computational requirements are overwhelming.

1.3.1 Generation Expansion Plans

Utilities can either build up new generation units or improve the efficiency of currently existing generation units to satisfy the increasing demand. However, in this thesis, we only consider the case of building up new generation units. For currently existing generation units, we need to determine their production levels year by year.

A set of projects are proposed by the people who are involved in the issues of designing new generation units. The potential generation units are different in their generation types and their locations. Each potential generation unit may have a "time window" indicating the possible time for its construction. Therefore, there are lots of projects. When these projects are proposed, the project characteristics such as type, expected fixed cost and the data for estimating its production cost must be predicted. The decision makers of utilities should select the projects with the minimum cost while satisfying a set of constraints. Moreover, at most one project can be selected for each potential generation unit over the whole planning

horizon.

In addition to project selection, we need to determine the production levels of currently existing generation units and the production levels of the selected projects. The major features of generation expansion planning are that the future increases in demand over the planning horizon must be met through generation expansion.

1.3.2 A Generation System

For all types of generation units, there are mainly two types of costs associated with each generation unit: fixed cost and production cost. Fixed costs are the costs for taxes, insurance, and maintenance. Production costs are the expenses with the operation of generation such as fuel costs. However, sometimes operating cost is used instead of production cost. Beglari and Laughton [1975] define operating costs of a plant as those expenses which are incurred only when the plant is run.

However, the costs associated with a generation system in the literature are generally relatively easy to be quantified, there are several types of costs that are seldom considered in generation expansion planning studies, such as the costs of outage, pollution, and competition. The reasons are probably because they are relatively hard to be quantified or the related studies are still not satisfied. In this thesis we will include outage cost in the model formulation.

In addition, we only consider generation systems in the cost minimization problems. in this thesis. Transmission and distribution planning considerations are assumed to be independent of expansion planning. This assumption is also made in many other expansion planning studies (see, Ammons [1981]).

1.3.3 Capital Budgeting Considerations

Capital budgeting is defined as the process of selecting capital investments in Levary and Seitz [1990]. Because the money required for the investments in utilities is generally enormous, the capital budgeting process for generation expansion planning deserves high attention from decision makers.

The capital budgeting process at utilities can be divided into several steps. First, the people who are charged for the future expansion planning of utilities should prepare projects that satisfy the requirements that are restricted by utilities. Second, the selection of projects will be done according to the results of optimization process by inputing the data of projects to mathematical models. However, the more accuracy we get during these processes, the more convincing our final project selections are. Moreover, because the planning horizon may last for more than thirty years, the impact of inflation must be included in capital budgeting analysis. In addition, to accurately reflect the true costs of electric power operations, easy-to-quantify costs such as fixed costs and operation costs and hard-to-quantify costs such as outage costs should be considered.

1.4 Least Cost Investment

In recent years the rapid change in the power environment has brought the utilities to invest more concerns in the capital budgeting process. Also, followed by a more competitive environment, the deregulation has altered the traditional economic behavior existing among utilities to a more cost effectiveness oriented situation. To survive in present power market, utilities really need to find more efficient and effective ways to design generation plans to satisfy customer demand.

Because the money needed in the electricity supply industry is extraordinary huge, operations research techniques were first applied to the operations in electric utilities. Anderson [1972] has a review paper which lists the commonly used model types for determining least-cost investments such as linear, nonlinear, dynamic programming models. By using these optimization techniques, utilities can determine the least-cost investments with theoretical bases.

1.5 Demand-side Management

To incorporate the continuously increasing demand for electric power, utilities had invested an enormous amount of money in constructing new power plants. For the construction of new power plants, not only the immense sums of investments are a heavy burden for utilities, but

also the long period of constructing time will place utilities under highly risky circumstances. Because the rapidly changing environment of the electric power industry is sometimes hard to forecast, it is likely that utilities might not be able to earn all their investments back. In short, the traditional supply-side management process is appropriate as long as energy resources are cheap and plentiful. However, we know the natural resources of the earth are limited and the cost of electrical power service is not cheap. So, in order to be able to supply the ever-increasing demand for electric power, utilities and the commissions of federal and state governments suggest that demand-side management be an alternative way to benefit both utilities and customers.

Demand-side management is the strategy searching for the shift or reduction of the demand for electric power. For example, utilities may use pricing strategies to induce customers to shift more of their electric power use from peak hours to off-peak hours. Demand-side management program is a strategy of satisfying the demand of customers in the ways of altering the demands of customer that will be better for both utilities and customers. Bartsch and Devaul [1993] says, "demand-side management has become a code phrase for an amalgam of activities that seek to control the costs of providing power. Rather than focus solely on their supply side by building new power plants, utilities and state regulators increasingly have realized the benefits of adjusting their customers' demand for electricity." Also, Pirkey and Millhone [1989] illustrate, "the Energy Information Administration projects that there will be a gap of 53 gigawatts (GW) in the year 2000 between expected demand and likely capacity. Option for helping to meet that need is the use of demand-side management strategies."

As more utilities have adopted demand-side management strategies and become more familiar with them, almost all customer sectors are included in demand-side management programs. In the residential and commercial sectors, the primary sources of potential power savings come from cooling, heating, and refrigeration. For instance, the commercial sector can use cool storage technologies to shift their load. In the industrial sector, the major saving potentials are in production processes. Briefly, by moving energy-intensive operations from the peak hours of energy use to off-peak hours is a common way for load shifting. However, the issue for

customers to be in demand-side management is to find and join the demand-side management programs provided by utilities that minimize the total cost of customers.

As mentioned previously, most portions of this country will be short of energy in the near future. To overcome this situation, utilities can either rapidly construct new supplies or implement efficient managements on the demand side. Bartsch and Devaul say, "If utilities can make investments that save money instead of building new energy-supply facilities, customers in their service territory are better off. A variety of regulatory procedures can make those investments profitable for both utility and customer, as well as for the community as a whole."

1.6 Nature of Integrated Resource Planning

Generation expansion planning only considered supply options. All the projects in capital budgeting process were designed to satisfy a given demand. However, since demand-side management has become another option of utilities for meeting the customer demand, integrated resource planning has drawn considerably attention from both the government and the public. Kreith [1993] says, "The National Energy Policy Act of 1992, signed by Ex-President Bush on October 24, 1992, attempts to improve the efficiency of energy use by various means, including a requirement that all utilities shall employ integrated resource planning." However, the term integrated resource planning (IRP) has been defined in several ways. The definition in the Act is "a planning and selection process for new energy resources that evaluates the full range of alternatives, including new generating capacity, power purchases, energy conservation and efficiency, cogeneration and district heating and cooling applications, and renewable energy resources to provide adequate and reliable service to its (the utility's) customers at the lowest system cost." Moreover, Kreith [1993] states, "IRP shall treat demand and supply resources on a consistent and integrated basis". Hoog and Hoobs [1993] defines, "IRP is the process of finding the optimal combination of supply-side and demand-side measures to meet the region's electricity needs, and is an area of increasing focus for electric utilities throughout the world."

As stated in Kreith [1993], there is no single method for integrated resource planning. However, Kreith [1993] has the following steps for the implementation of integrated resource

planning.

1. Develop a load forecast.
2. Inventory existing resources.
3. Identify future electricity needs not being met by existing resources.
4. Identify potential resource options, including demand-side management programs.
5. Screen all options to identify those that are feasible and economic.
6. Identify and quantify environmental and social costs of these options.
7. Perform an uncertainty analysis.
8. Select a preferred mix of resources.
9. Implement least cost mix of supply and conservation options.

In this thesis, we will formulate a comprehensive integrated resource planning model where the load forecasts are assumed to be perfectly reliable. In addition, environmental costs are not considered in our model. Instead, we focus on the quantification of outage cost and we will include them in an integrated resource planning model.

1.7 Role of Outage Costs

We divide power plant characteristics impacted by capital projects into two groups. One is relatively easy-to-quantify factors such as production cost and fixed cost. These factors are expressed in or easily transformed to monetary terms. (see, e.g., Skugge and Jabbour [1995]). The other is relatively hard-to-quantify factors such as pollution and outage costs. Compared to easy-to-quantify factors, these factors are not easily transformed to monetary values.

Generation expansion planning has received considerable attention, and many mathematical models have been developed up to date. However, the models in the literature mainly focus on easy-to-quantify factors. Some of the models in the literature consider outage cost in the form of reliability because the close relation between outage cost and reliability.

1.7.1 Definition of Outage and Reliability

According the causes of outages' occurrences, there are two types of outages. The definitions of both outages in the IEEE standard dictionary of electrical and electronics terms [1988] are as follows.

Forced outage: A power outage that results from the failure of a system component, requiring that it be taken out of service immediately, either automatically or by manual switching operations, or an outage caused by improper operation of equipment or human error. This type of power outage is not directly controllable and is usually unexpected.

Scheduled outage: A loss of electric power that results when a component is deliberately taken out of service at a selected time, usually for purposes of construction, preventive maintenance, or repair. The key test to determine if an outage should be classified as forced or scheduled is as follows. If it is possible to defer the outage when such deferment is desirable, the outage is a scheduled outage, otherwise, the outage is a forced outage.

In this thesis, only forced outage will be discussed together with our integrated resource planning model.

Burns and Gross [1990] claims, "the basic approach to measuring reliability worth is in terms of customer outage costs." The definition of reliability in IEEE standard dictionary of electrical and electronics terms [1988] is as follows.

Reliability: The ability of an item to perform a required function under stated conditions for a stated period of time.

1.7.2 Outage and Reliability

Traditionally, utilities attempt to construct new generation units to meet the increased customer demand for power. Reliability of power supply has been thought as a critical issue for utilities. However, because the stochastic characteristics of forced outage, there is always a probability that customer demand may not be met. We will discuss this observation in details in Chapter 3.

Two reliability planning criteria are broadly used in the mathematical models for generation

expansion planning. One is reserve margin constraint (i.e., please refer to constraint (3.3) in Chapter 3). The other is the evaluation of loss of load probability which is a probabilistic criteria (i.e., please refer to constraint (2.5) in Chapter 2).

1.8 Summary

The contributions of this thesis mainly lie in the area of model formulations. The model developed in this thesis includes utility's outage costs and some factors that reflect the new environment of the present electric industry.

Before demand-side management techniques were broadly used, utilities only considered supply options to meet the electric power demand of customers. In order to supply more power to meet the ever-increasing demand of customers, utilities have continuously expanded their generation capacity. The techniques are named as generation expansion planning and we review the relevant research in Chapter 2. However, because the constructions of new generation units are so costly and the natural resources for the needs of electric power are limited, utilities are starting to think about the options of reducing customer demand or altering the timing of customer demand. These new strategies are called demand-side management and have been implemented in most of the states in this country. So, some mathematical models for least cost planning that combine demand-side management with the supply-side management have been seen in the literature. These techniques are called integrated resource planning. In the later part of Chapter 2 we review integrated resource planning research that is different from generation expansion planning in their concepts of how to meet customer demands. Model development in Chapter 3 combines the concepts of generation expansion planning and integrated resource planning. The model is more like an integrated resource planning model, and includes capital budgeting process. In other words, project selection is considered in the model. In addition, the model includes quantified outage costs which are said one of the hard-to-quantify factors in the capital budgeting process of electric power utilities. The details of the quantification of outage costs are presented in Chapter 4.

Because the model developed in this thesis is a mixed integer linear programming problem.

Integer programming problems are generally more difficult to be solved than linear programming problems. Therefore, we introduce the algorithm for solving mixed integer linear programming problems and an applicable software package in Chapter 5. Finally, we conclude this thesis by summarizing what we did and the potential research areas for the future in Chapter 6.

2 REVIEW OF RELEVANT LITERATURE

2.1 Introduction

In this chapter we would like to review the relevant literature for generation expansion planning, as well as integrated resource planning. As stated before, operation research techniques have been extensively applied for determining the least cost investments for utilities. Anderson [1972] has a taxonomy paper introducing the commonly used optimization techniques. Linear, nonlinear, and dynamic programming optimization models are reviewed in his paper. Also, simulation models are discussed in his paper because some factors may alter over the planning horizon and have specific statistical distributions. However, the models reviewed in his paper are illustrated and extended by a large number of later authors. Irisari [1975] presents a dynamic programming optimization model for power system expansion planning. A long-range generation planning model is developed in Bloom [1982] and several methodologies for solving this model are introduced.

Ammons and McGinnis [1985] develop a generation expansion planning model which will be used as the element reference of this thesis and will be discussed together with an integrated resource planning model developed by Hobbs [1995a]. A stochastic optimization model for a generation system with the consideration of unserved demand is stated in Hobbs [1993]. However, because of the pass of the 1990 U.S. Clean Air Act Amendments, the environmental concerns are included in some generation expansion planning models. Hobbs [1994] presents a generation model that chooses the operation level for each generation unit such that the total system cost is minimized. A feature of his model is that the model reflects the environmental concerns which have become critical factors in recent years. Similarly, we can find the impacts of environmental policies on utility planning and operations in Hobbs and Centolella [1995c].

Besides the formulation of generation expansion planning models, some authors combine transmission systems and generation systems and formulate mathematical models for determining least-cost investments. For example, Sawey and Zinn [1977] develop a linear programming model for long range expansion planning of generation and transmission systems. This model provides construction schedules for transmission lines and power plants. Similarly, Beglari and Laughton [1975] formulate three models to determine investment plans. One is a linear programming model designed for the economic planning of generation. One is formulated for minimizing the total present-valued costs of transmission systems. The other model is designed for minimizing the total cost of the combination of generation and transmission systems.

At the same time, reliability considerations draw lots of attention in mathematical formulations. Generally, a generation expansion planning problem will become more complicated and hard to solve after we include the concern of reliability. Reserve margin and loss-of-load-probability constraints are most commonly used among the generation expansion planning models. A few papers have discussed how to handle reliability constraints in the optimization of generation expansion planning. Sanghvi and Shavel [1984] have a review paper introducing the commonly used methodologies for incorporating reliability constraints in optimization models for generation expansion planning. More recently, Niimura and Kermanshahi [1990] presents the use of loss of load probability as an index showing supply reliability. In their paper, a multi-stage optimization procedure with the calculation of loss of load probability of present generation units plus the determination of loss of load-probability of future generation units are presented. Also, loss of load probability is discussed in detail in Ammons's [1981] and Irisari's [1975] dissertations.

Incorporating the change of the power industry, integrated resource planning has attracted much of attention from scholars in recent years. Hirst and Goldman [1991] and Kreith [1993] state the recent progress in integrated resource planning research. As stated in Chapter 1, integrated resource planning treats demand and supply resources on a consistent and integrated basis (see e.g., Kreith [1993]). Hobbs et al. [1993] introduce the basic structure of an integrated resource planning model that minimizes the total cost of a utility. Hoog and Hobbs [1993]

examine what are needed for a comprehensive model for integrated resource planning. However, Hobbs [1995a] has a mixed-integer linear resource planning model which is also used as an important reference in this thesis. In the later sections of this chapter, we mainly focus on the explanation of the mathematical models in Hobbs [1995a] and Ammons and McGinnis [1985] because they provide us some basis for an improved mathematical model for capital budgeting process. The comprehensive comparison among the model in Hobbs [1995a], the model in Ammons and McGinnis [1985], and our model is provided in Chapter 5.

2.2 A Generation Expansion Planning Model by Ammons and McGinnis

Ammons and McGinnis [1985] formulate a comprehensive generation expansion planning model for electric utilities. Because a quadratic function for estimating production cost is included in the objective function, it is a mixed integer nonlinear programming problem. Moreover, from the features of this model, we recognize that it is developed from a typically traditional supply-side management view point. Because our basic model in Chapter 3 is based on the model in Ammons and McGinnis [1985], a version of their model is stated in next section.

2.2.1 The Model

Ammons and McGinnis [1985] has a mixed integer nonlinear programming model as follows:

$$\min_{x_{jit},y_j,S_t} \sum_{j=1}^{J} f_j y_j + \sum_{t=1}^{T}\sum_{i=1}^{I}\sum_{j=1}^{J} Q_{jit} H_{it} \tag{2.1}$$

subject to

$$\sum_{j=1}^{J} x_{jit} \geq P_{it} \tag{2.2}$$

$$L_{jt} y_j \leq x_{jit} \leq a_{jt} U_{jt} y_j \tag{2.3}$$

$$S_t - (1 + r_t) S_{t-1} + \sum_{j=1}^{J} E_{jt} y_j = B_t \tag{2.4}$$

$$LOLP_t(\{y_j\}) \leq LOLP_{t-max} \tag{2.5}$$

$$\sum_{j=1}^{J} a_{jt} U_{jt} y_j \geq (1+m) P_t \tag{2.6}$$

$$\sum_{j \in \{J_k\}} y_j \leq 1 \tag{2.7}$$

$$y_j \in \{0,1\} \tag{2.8}$$

$$S_t \geq 0 \tag{2.9}$$

$$Q_{jit} = \begin{cases} V_{1jt} x_{jit}^2 + V_{2jt} x_{jit} + V_{3jt} & \text{if } x_{jit} > 0, \\ 0 & \text{otherwise} \end{cases} \tag{2.10}$$

where

y_j = zero-one valued integer variable indicating whether project j is to be constructed

x_{jit} = operation level of generation unit j in interval i of period t

f_j = sum of discounted fixed costs associated with generation unit j over the planning horizon

$V_{.jt}$ = discounted coefficients of quadratic production cost function for generation unit j in period t

Q_{jit} = discounted production cost associated with project j in interval i of period t

H_{it} = number of hours in interval i of period t

P_{it} = power demand level in interval i of period t

L_{jt} = lower bound on production level on generation unit j in period t

U_{jt} = upper bound on production level on generation unit j in period t

a_{jt} = availability factor for generation unit j in period t, $0 \leq a_{jt} \leq 1$

m = portion of capacity reserve margin required when peak demand occurs, $m \geq 0$

P_t = peak demand during period t

S_t = construction budget funds unused in period t, $S_0 = 0$

r_t = short term rate of return for period t

E_{jt} = construction funds required for generation unit j in period t

B_t = additional construction budget funds available in period t

$LOLP_t(\{y_j\})$ = the loss of load probability of system $\{y_j\}$ during period t

$LOLP_{t-max}$ = the maximum allowable loss of load probability during period t

$\{J_k\}$ = set of indices belonging to mutually exclusive projects, e.g., projects associated with a single generation unit

However, the model presented here is slightly different from the original model in Ammons and McGinnis [1985]. For the physical constraints of the production level, we change the upper bound $U_{jt}y_j$ to $a_{jt}U_{jt}y_j$ as shown in constraint 2.3. As defined, a_{jt} is the availability factor which only refers to forced outage and is generally about 0.90. Because forced outage is not directly controllable and is generally unexpected, the upper bound $U_{jt}y_j$ for the production level in their model never can be reached. So, we suggest including the availability factor in the upper bound. However, a similar concern for multiplying capacity by the availability factor as the upper bound for the operation level can be seen in the literature (see, e.g., Anderson [1972]).

Financial considerations can be seen in the budget constraint (constraint 2.4). The characteristics of this model different from other generation expansion planning models are that project selection or capital budgeting concern is included. This feature can be reflected by the set of system configuration constraints (constraints 2.7 and 2.8). Over a long range planning horizon, a generation system may have a huge diversity of projects to be evaluated. Because project evaluations occur at year zero, discounted costs or present values are used. However, these capital budgeting processes require the techniques of engineering economy.

Reliability constraints that guarantee a certain production level in each time period are considered. Loss of load probability constraint (constraint 2.5) states that for every project the loss of load probability cannot be bigger than a maximum level which is restricted by a utility.

Another feature of this model is that they do not use a linear function of production level to estimate production costs. Instead, a quadratic function of production level is claimed as a good estimate for production costs. Because of this, the objective function becomes nonlinear and the whole model becomes a mixed integer nonlinear programming problem.

Two sets of demand satisfaction constraints are formulated. A set of demand constraints are developed to assure that customer demand is always satisfied (constraint 2.2). Moreover, a set of peak demand satisfaction constraints are developed to restrict that the peak customer demand plus a certain amount of reserve margin are satisfied (constraint 2.6). We may regard this set of constraints as reliability or regulation constraints because this restriction is sometimes regulated by governments. However, if we look at the implications of constraint (2.2) carefully, we might find that, in this model, utilities never produce less power than the expected customer demand. This phenomenon reveals that this model attempts to satisfy customer demand only from the traditional supply-side management. However, because of the popularity of demand-side management, to be more consistent with the present power industry, we might also need to consider the ways of satisfying customer demand with demand-side management. Therefore, the inspirations of reformulating this model to accurately reflect the present utility industry are sparkled.

2.3 A Resource Planning Model by Hobbs

In contrast to the model in Ammons and McGinnis [1985], Hobbs [1995a] includes the demand-side management and develops an integrated resource type model. Actually, the model in Ammons and McGinnis [1985] and the model in Hobbs [1995a] indicate two generation expansion planning concepts representing two eras of the electric industry. Before the 1990's, the planner's task was to determine the types of generation units to be constructed, the timing for these constructions, and the production level of every generation unit for new and existing units to meet growing customer demand. Now, however, utilities have more options to meet customer demand in the era of deregulation. For example, utilities adopt demand-side management strategies such as load control and combine with traditional supply-side man-

agement strategies in order to meet customer demand in Hobbs [1995a]. A solution to this model will determine which generation units and demand-side management programs should be constructed to minimize the sum of fixed, production, and demand-side management costs of a utility and outage costs of customers at the same time.

2.3.1 The Model

In this subsection, we state the model in Hobbs as follows:

$$\min_{x_{iy},d_{ky},u_{yt},g_{iyt}} \sum_{y=1}^{Y}\sum_{i=1}^{I} CX_{iy}x_{iy} + \sum_{y=1}^{Y}\sum_{t=1}^{T}\sum_{i=1}^{I} CG_{iyt}H_t g_{iyt} + \sum_{y=1}^{Y}\sum_{k=1}^{K} CD_{ky}d_{ky} + \sum_{y=1}^{Y}\sum_{t=1}^{T} CU_{yt}H_t u_{yt} \tag{2.11}$$

subject to

$$\sum_{i=1}^{I} g_{iyt} + \sum_{k=1}^{K}\sum_{v\le y}^{V} SAV_{k,t,y-v}d_{kv} + u_{yt} = LOAD_{yt} \quad \forall y,t; \tag{2.12}$$

$$\sum_{i=1}^{I}\sum_{v\le y}^{V} x_{iv} + (1+RM)\sum_{k=1}^{K}\sum_{v\le y}^{V} SAV_{k,1,y-v}d_{kv} \ge (1+RM)LOAD_{y,1} \quad \forall y; \tag{2.13}$$

$$g_{iyt} \le (1-FOR_i)\sum_{v\le y}^{V} x_{iv} \quad \forall i,y,t; \tag{2.14}$$

$$\sum_{t=1}^{T} H_t g_{iyt} \le 8670 MCF_i \sum_{v\le y}^{V} x_{iv} \quad \forall i,y; \tag{2.15}$$

$$\sum_{v\in V_k} d_{kv} \le 1 \quad \forall k; \tag{2.16}$$

where the model's decision variables are:

x_{iy} = generation capacity (mw) of generation unit i constructed in year y.

g_{iyt} = generation level (mw) during interval t of period (year) y from generation unit i.

d_{kv} = a zero-one variable indicating whether demand-side management program k has been implemented beginning in y.

u_{yt} = the amount of demand (mw) that goes unserved in interval t of period y.

where the model's parameters are:

CD_{ky} = the present worth of the cost ($) of implementing demand-side management program k starting in period y.

CG_{iyt} = the present worth of the variable operating and maintenance cost ($/mwh) of generation unit i during y and t.

CU_{yt} = the present worth of the cost ($/mwh) to consumers of interrupting power supplies during y and t.

CX_{iy} = the present worth of the construction and fixed cost ($/mwh) resulting from building a mw of capacity of generation unit i that first becomes available in period y.

FOR_i = probability of generating capacity i being unavailable due to the unscheduled outages (forced outage rate).

H_t = length (hours) of interval t. $\sum_{t=1}^{T} H_t$ equals 8760, the number of hours in a year.

$LOAD_{yt}$ = electricity demanded (MW), including transmission and distribution losses, during y and t, if no demand-side management programs are implemented. The peak demand during the year is assumed to occur during $t = 1$

MCF_i = maximum annual capacity factor (average power output/capacity) for generation unit i, considering maintenance and forced outages.

RM = capacity reserve margin (excess of capacity over peak demand required to meet reliability needs).

$SAV_{k,t,y-v}$ = decrease in demand (mw) during y and t resulting from implementation of demand-side management program k which starts from year v and has been implemented for $y - v$ years.

V_k = the set of indices indicating all possible commissioning periods of construction initiation for the same demand-side management program k.

Again, the model presented here is slightly different from the original model in Hobbs [1995a]. Constraint (2.16) is added because we need to have a mutually exclusive restriction for demand-side management programs. More specifically, from the notations shown above, we note that d_{kv} is "a zero-one variable indicating whether demand-side management program

k has been implemented beginning in v". However, to make the model in Hobbs [1995a] more reasonable, we include constraints (2.16) to restrict that each demand-side management program can be chosen at most once over its possible commissioning periods. In this model, once the selected demand-side management program starts, it never ends over the whole planning horizon.

Some special characteristics are presented in this model. Because demand-side management is considered, customer demand is not regarded as "given" information, but it can be partly saved by adopting demand-side management programs. As shown in constraints (2.12) and (2.13), demand-side management may save some of customer demand. Attempting to satisfy customer demand from both demand-side and supply-side management is a special feature of this model.

Another feature of this model is that the amount of unserved demand is one of the decision variables. This concept can be seen in Hobbs's papers (see, e.g., [1993b], [1994], [1995b]). Like we explained in Chapter 1, there are interruptible and curtailable programs that exist in the utility industry. In Hobbs's models, utilities can determine the level of unserved demand (i.e., scheduled outage). On the other hand, forced outage is also included in the constraints of this model. In this model forced outage is not expressed as loss of load probability which has been broadly used in the literature. Instead, a parameter called FOR_i is used in constraints (2.14) and Hobbs claims that "it is an attempt to include the effect of random outages upon generator output within a deterministic mathematical program".

In addition, from constraints (2.12) we know that utilities always produce less power than the expected customer demand. The interpretations for constraints (2.12) are that customer demand before any implementation of demand-side management is always greater than or equal to the generation level. More specifically, utilities are not allowed to produce more power than the expected customer demand. However, in the real world, utilities can sell their extra power to other utilities. Therefore, we claim that the integrated resource model in Hobbs [1995a] does not quite fit the current utility environment and should be slightly changed.

We will develop our integrated resource planning model in Chapter 3. As we mentioned

earlier, a comprehensive comparison among the model in Ammons and McGinnis [1985], the model in Hobbs [1995a], and our model is provided in Chapter 5.

3 INTEGRATED RESOURCE PLANNING: BASIC MODEL

3.1 Introduction

In this chapter, we construct our basic model of integrated resource planning for a utility. The objective is the minimization of the total costs, and the constraints include the budget consideration, the supply and demand relations, and reliability restrictions.

3.2 Decision Variables

In our model, there are five types of decision variables. Our primary task is to find the optimal values of these five types of decision variables that minimize the total costs. By obtaining these values, we determine which new projects to be selected. At the same time, we determine the optimal operational production levels of the existing generation units and the selected new projects (generation units) as well. In addition, we determine the amount of power saved by demand-side management programs and the amount of unused budget funds. Therefore, by solving the basic model, we can determine both the supply-side and the demand-side management strategies of a utility. Let us now explain decision variables.

Let y_j denote the set of binary decision variables. j is the index of generation expansion planning projects and currently existing generation units. The proposed projects may be differentiated by their times of commissioning, types of new generation units, and their sizes. In other words, each project represents a particular generation unit that can be constructed (chosen). A value of zero for y_j means that project j is not to be constructed. A value of one for y_j means that project j is to be constructed. For those currently existing generation units, the corresponding y_js are pre-fixed to be one. That is, no decision will be made on the existing generation units via these binary decision variables. Therefore, our model cannot only

determine which new generation units should be constructed, but also the production levels of existing and new generation units over the planning horizon.

A second set of decision variables is x_{jit}, representing the production level of generation unit j in interval i of period t. We need to determine x_{jit} for both existing and new generation units that are to be constructed. Moreover, since we want to allow both power surplus and power deficit, x_{jit} consists of two kinds of variables b_{jit} and d_{jit}. By b_{jit}, we denote the amount of power generated and delivered to customers by generation unit j in interval i of period t. By d_{jit}, on the other hand, we denote the amount of power generated but not delivered to customers by generation unit j in interval i of period t. The details of power deficit and power surplus will be discussed in section 3.4.

A third set of decision variables is S_t, representing the construction budget funds unused in period t. We formulate a set of capital budgeting constraints by combining this variable with the construction costs for new generation units, the details will be discussed in the next section.

The last set of decision variables is u_{it}, representing the amount of power saved by demand-side management strategies in interval i of period t. In practice, there are interruptible and curtailable programs in the utility industry. Customers can get price discounts in exchange for the agreements of reducing their demands from utilities. Customers have an incentive to participate in such a program if the expected benefits from the program exceed the expected costs. For utilities, how to determine priority services for those participants is crucial, but it is beyond the extent of this thesis.

Thus far we have explained the decision variables of our model. We now formally present the definitions of these decision variables.

Definitions:

y_j=binary variable indicating whether project j is to be constructed ($y_j = 1$ implies construction, $y_j = 0$ implies no construction)

b_{jit}=power generated and delivered to customers by generation unit j in interval i of period t

d_{jit}=power generated but not delivered to customers by generation unit j in interval i of period t (e.g., reserve margin)

x_{jit}=power generated (production level) by generation unit j in interval i of period t which is the sum of b_{jit} and d_{jit}

S_t=construction budget funds unused in period t, $S_0 = 0$

u_{it}=the amount of power saved by demand-side management programs in interval i of period t

3.3 Objective Function

The coefficient associated with the binary variable y_j in the objective function is the discounted fixed cost f_j. For each j, there is a corresponding discounted fixed cost f_j for the expenses of taxes, insurance, depreciation and maintenance. For the proposed projects, this cost occurs only when the corresponding project j is selected or when the corresponding y_j is equal to one. For existing generation units, fixed cost always exists because the corresponding y_js are pre-fixed to be one. Because integrated resource planning model is for solving large-scale planning problems which may span twenty to thirty years, the considerations of time value of money is important in capital budgeting process. In this model, all costs are expressed in the present worth.

Importantly, for the reasons of computationally tractible ease, especially for large-scale planning problems, we choose linear form for the production cost function. The same assumptions can be seen in the literature (see, e.g., Anderson [1972], Hobbs and Centolella [1995], and Hobbs [1995a]). Here, we use notation $F_{jit}x_{jit}$ to indicate hourly production cost associated with project j in interval i of period t, where F_{jit} is the discounted production cost per unit of energy output for generation unit j in interval i period t.

Another coefficient is H_{it}, representing the number of hours in interval i of period t. Each time period represents a cycle that repeats the same pattern of customer demand over the planning horizon. A time period generally consists of peak and off-peak hours (periods).

The coefficient associated with decision variable u_{it} is the demand-side management (DSM)

cost for one unit energy saved for a utility in interval i of period t, denoted by SO_{it} (\$/kwh). As stated in Kreith [1993], demand-side management cost for one unit energy saved can be expressed as

$$\frac{initial\ cost\ of\ DSM\ program * CRF}{energy\ saved\ per\ year}(\$/kwh)$$

where CRF is the capital recovery factor that accounts for the time value of money invested initially and can be expressed as

$$CRF(r,t) = \frac{r}{1-(1+r)^{-t}}$$

where t is the lifetime of the demand-side management program and r is the discount rate. Kreith [1993] interprets $CRF(r,t)$ as "the ratio of the annual payments to the total sum that must be repaid". However, because there exists many demand-side management programs, in our model SO_{it} can be viewed as the averaged cost over all possible programs. It might be different from time to time because of inflation rate or the change in demand-side management program selections. Finally, we can interpret $SO_{it}u_{it}H_{it}$ as the cost of implementing demand-side management programs in interval i of period t.

We express the forced outage costs to a utility as $FOC_{it}FO_{it}H_{it}$ and it can be thought as the revenue loss to a utility. FOC_{it} is the forced outage cost for one kwh outage in interval i of period t and can be thought as the averaged price of electricity over all types of the customers of a utility. FO_{it} is the expected amount of forced outage in interval i of period t. Due to the stochastic features of forced outage, Ammons and McGinnis [1981], Irisari [1975], Ghajar and Billinton [1993], Niimura and Kermanshahi [1990], and Bloom [1982] have demonstrated that the expected amount of forced outage can be readily computed from the available capacity and demand distributions. They have presented that the levels of customer demand and available capacity will determine the expected amount of forced outage. From this basis, we construct a regression model consisting of the expected amount of forced outage as the dependent variable and the levels of customer demand and available capacity of a utility as the independent variables. Namely,

$$FO_{it} = \alpha\sum_{j=1}^{J} a_{jt}U_{jt}y_j + \beta P_{it} + \gamma$$

The definitions for this linear regression model are as follows.

Definitions:

FO_{it} = the expected amount of forced outage in interval i of period t (mw)

a_{jt} = availability factor for generation unit j in period t, $0 \leq a_{jt} \leq 1$

U_{jt} = upper bound on production level (capacity) on generation unit j in period t (mw)

$\sum_{j=1}^{J} a_{jt} U_{jt} y_j$ = the expected totally available capacity in period t (mw)

P_{it} = the expected customer demand prior to the implementation of demand-side management in interval i of period t (mw)

α, β, and γ = the coefficients need to be estimated

Also, let us summarize the additional definitions used up to here as follows. Some of the definitions are similar to Ammons and McGinnis [1985].

Definitions:

f_j = sum of discounted fixed costs associated with generation unit j over the planning horizon

F_{jit} = discounted production cost per unit of energy output associated with project j in interval i of period t (\$/mwh)

H_{it} = number of hours in interval i of period t

SO_{it} = the demand-side management cost for one unit energy saved in interval i of period t (\$/mwh)

FOC_{it} = the forced outage cost per mwh to utilities (e.g., loss of revenue) in interval i of period t (\$/mwh)

The objective is to minimize the sum of discounted fixed, production, demand-side management, and forced outage costs for a utility. Hence, the objective function has the form

$$\min_{b_{jit}, d_{jit}, y_j, S_t, u_{it}} \sum_{j=1}^{J} f_j y_j + \sum_{t=1}^{T} \sum_{i=1}^{I} \sum_{j=1}^{J} F_{jit} x_{jit} H_{it} + \sum_{t=1}^{T} \sum_{i=1}^{I} SO_{it} u_{it} H_{it} + \sum_{t=1}^{T} \sum_{i=1}^{I} FOC_{it} FO_{it} H_{it}$$

3.4 Constraints

Let us first examine the relations among power production quantities x_{jit}, b_{jit}, and d_{jit} (Recall x_{jit}: power generated by generation unit j in interval i of period t, b_{jit}: power generated and delivered to customers by generation unit j in interval i of period t, d_{jit}: power generated but not delivered to customers by generation unit j in interval i of period t). For clarity, we do not substitute x_{jit} with b_{jit} and d_{jit} in the mathematical expressions in this model. Production level is as follows.

$$x_{jit} = b_{jit} + d_{jit} \qquad for\ all\ i,\ j,\ and\ t \tag{3.1}$$

Next, let us examine the relations among b_{jit}, u_{it}, and P_{it} (Recall u_{it}: the amount of power saved by demand-side management in interval i of period t, P_{it}: the expected customer demand prior to the implementation of demand-side management in interval i of period t) Because demand-side management is considered in the model, there are two ways to meet customer demand in interval i of period t. One is the sum of power generated and delivered to customers by all generation units in interval i of period t (i.e., $\sum_{j=1}^{J} b_{jit}$). The other is the amount of electric power demand saved by demand-side management programs in interval i of period t (i.e., u_{it}). Therefore, there exists a power equation as follows.

$$\sum_{j=1}^{J} b_{jit} + u_{it} = P_{it} \qquad for\ all\ i\ and\ t \tag{3.2}$$

Hence equation (3.2) states that the sum of the power delivered to customers plus the power saved via demand-side management is equal to the expected customer demand prior to the implementation of demand-side management programs. The main feature of integrated resource planning is shown in this set of constraints. That is, utilities attempt to satisfy customer demand from both demand-side and supply-side management.

Another feature of our model is the consideration of power surplus and power deficit that can be observed in (3.1) and (3.2). For instance, power surplus can be observed if $\sum_{j=1}^{J} d_{jit}$ is greater than zero and u_{it} is equal to zero. On the other hand, power deficit can be observed if $\sum_{j=1}^{J} d_{jit}$ is equal to zero and u_{it} is greater than zero. Power deficit and power surplus are explicitly defined as follows.

Power deficit: If the amount of power generated by a utility is less than the customer demand prior to demand-side management programs , we say that there exists power deficit. To eliminate this deficit, a utility can either implement demand-side management programs (e.g., scheduled outage) to modify the timing and amount of power demand or purchase power from other utilities.

Power surplus: If the amount of power generated by a utility is more than customer demand prior to demand-side management programs, we say that there exists power surplus. For this extra power, a utility can sell it to other utilities.

Importantly, because of the random nature of forced outage (e.g., unexpected failure and repair of a generation unit), forced outage may occur in any circumstances (i.e., whether we expect power deficit or power surplus). Moreover, we think the considerations of power deficit and surplus are important because outside purchase and outside sale for power have been common economic activities among utilities for several years.

Also, no generation unit can produce more power than its available capacity. Likewise, no generation unit can produce less power than its minimum power generating level (Ref., Anderson [1972], Ammons and McGinnis [1985]). Namely,

$$L_{jt}y_j \leq x_{jit} \leq a_{jt}U_{jt}y_j \quad \textit{for all } i,\ j,\ \textit{and } t$$

we note that a_{jt} is the availability factor of generation unit j (i.e., existing and potential generation units) in period t. This factor refers only to forced outage (and not to scheduled outage) and can be obtained by using the past performance record of the similar types of generation units. U_{jt} is the upper bound (capacity) on production level. L_{jt} is the lower bound on production level. We note that this constraint is non-trivial only if $y_j = 1$ (i.e., the generation unit j is chosen for construction or is a pre-existing unit.). However, a potential generation unit that will be constructed does not generate any power during construction.

In addition, there exist budget constraints. They are modeled as

$$S_t - (1 + r_t)S_{t-1} + \sum_{j=1}^{J} E_{jt}y_j = B_t \quad \textit{for all } t$$

where B_t indicates the construction budget funds available in period t. E_{jt} is the construction funds required for constructing potential generation unit j in period t. As stated before, the value of one of y_j indicates that either project j is to be constructed or it is an existing generation unit. In the case of constructing potential generation units, E_{jt} is fixed to be zero before the start of construction and after the end of construction (i.e., this information is provided by project designers). In the case of existing generation unit, all E_{jt} are equal to zero because no construction funds are required. r_t means rate of return. S_t represents the construction funds unused in period t (i.e., S_t is the remaining amount after the expenditures for period 1 through t. we assume S_0 is zero).

Moreover, reliability constraints have been commonly used in generation expansion planning. Here, we use reserve margin, m, in characterizing the reliability constraint. It can be written as

$$\sum_{j=1}^{J} a_{jt} U_{jt} y_j \geq (1+m) P_t \qquad for\ all\ t \tag{3.3}$$

where P_t is peak demand in period t and can be thought as the maximum amount of demand among all intervals in a same period. m is the reserve margin for extra demand above P_t.

We also have constraints on our binary variable y_j. y_j is the binary variable indicating whether or not a project j is to be constructed, and y_j is fixed to be one for existing generation units in the system. However, a particular potential generation unit may have several mutually exclusive project variables y_j's, one for each possible commissioning year for construction over the whole planning horizon. If a set J_k contains the indices of the projects indicating all possible commissioning years of construction initiation for the same potential generation unit k, the following constraint allows at most one project for a specific potential generation unit to be constructed.

$$\sum_{j \in \{J_k\}} y_j \leq 1 \qquad for\ all\ k$$

In addition, the values of y_j should be zero or one. y_j is restricted with

$$y_j \in \{0, 1\} \qquad for\ all\ j$$

Finally, we have the following nonnegative constraints:

$$b_{jit} \geq 0, d_{jit} \geq 0, S_t \geq 0, u_{it} \geq 0 \quad \textit{for all } i,\ j, \textit{ and } t$$

After we explain the constraints of our model, now let us summarize the additional definitions used here as follows. As before, some of the definitions are similar to Ammons and McGinnis [1985].

Definitions:

L_{jt} = lower bound on production level on generation unit j in period t (mw)

r_t = short term rate of return for period t

E_{jt} = construction funds required for constructing potential generation unit j in period t

B_t = construction budget funds available in period t

P_t = peak demand during period t (mw)

m = the reserve margin for extra demand above P_t, $m \geq 0$

J_k = the set of indices of the projects indicating all possible commissioning years of construction initiation for the same potential generation unit k

3.5 Outage and Reliability

Theoretically, utilities may prevent any occurrence of forced outages by establishing a strict reliability standard in the supply of power. However, in real world, there is always a probability that customer demand may not be met, no matter how many generation units are built. That is because of the stochastic features of forced outage. We demonstrate this fact as follows.

Observation 3.1 Even though we actually have a strict, high reliability level (i.e., the available capacity is strictly greater than the expected customer demand) , forced outage still may occur.

Since both peak demand plus a certain amount of reserve margin as well as available capacity are stochastic, we assume, for simplicity, that they are uniformly distributed. The random variable x (peak demand plus reserve margin) is assumed to have a uniform density

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a < x < b, \\ 0 & \text{otherwise} \end{cases} \tag{3.4}$$

The random variable y (sum of available capacity) is assumed to have a uniform density

$$f(y) = \begin{cases} \frac{1}{d-c} & \text{for } c < y < d, \\ 0 & \text{otherwise} \end{cases} \tag{3.5}$$

In the constraint (3.3), expected values are used. Under the condition of constraint (3.3) being satisfied, the expected value of the sum of available capacity is greater than or equal to the peak demand plus a certain amount of reserve margin. Hence, from (3.4) and (3.5), we know that $\frac{a+b}{2} \leq \frac{c+d}{2}$. In addition, for the computational convenience, we assume that $c < b$ and $a < d$. Under these assumptions, we consider the case of $a < c < b < d$ only. We choose this specific case to show $Prob(forced\,outage) > 0$ under the condition of reliability constraint being satisfied.

We assume that x and y are statistically independent (i.e., the similar assumption can be seen in Ammons [1981] and Irisari [1975]). Then we can calculate Prob(forced outage) as the follows.

$$\begin{aligned} Prob(forced\,outage) &= Prob(y \leq x) \\ &= \int_c^b \int_y^b \frac{1}{(b-a)} \frac{1}{(d-c)} \, dx dy \\ &= \frac{(b-c)^2}{2(b-a)(d-c)} \end{aligned} \tag{3.6}$$

From (3.6), we know that Prob(forced outage) is always greater than zero. Therefore, the probability of the occurrence of forced outage always exists.

Now, we provide a small numerical example.

$$a = 100mw, b = 106mw, c = 105mw, d = 107mw$$

So, the expected value of peak demand plus reserve margin is 103 mw, which is less than the expected value of the sum of available capacity which equals 106 mw. By using formula (3.6), we get $Prob(forced\,outage) = 4.2\%$.

We reflect Observation 3.1 in our basic model. That is, we have the reserve margin as reliability constraints while forced outage cost as a component of the objective function.

3.6 General Model

Using the notation defined above, a mixed integer linear model combining demand-side and supply-side measures with the utility's outage costs is formulated as follows:

$$\min_{b_{jit},d_{jit},y_j,S_t,u_{it}} \sum_{j=1}^{J} f_j y_j + \sum_{t=1}^{T}\sum_{i=1}^{I}\sum_{j=1}^{J} F_{jit} x_{jit} H_{it} + \sum_{t=1}^{T}\sum_{i=1}^{I} SO_{it} u_{it} H_{it} + \sum_{t=1}^{T}\sum_{i=1}^{I} FOC_{it} FO_{it} H_{it} \quad (3.7)$$

subject to

$$\sum_{j=1}^{J} b_{jit} + u_{it} = P_{it} \quad (3.8)$$

$$x_{jit} = b_{jit} + d_{jit} \quad (3.9)$$

$$L_{jt} y_j \le x_{jit} \le a_{jt} U_{jt} y_j \quad (3.10)$$

$$S_t - (1 + r_t) S_{t-1} + \sum_{j=1}^{J} E_{jt} y_j = B_t \quad (3.11)$$

$$\sum_{j=1}^{J} a_{jt} U_{jt} y_j \ge (1 + m) P_t \quad (3.12)$$

$$\sum_{j \in \{J_k\}} y_j \le 1 \quad (3.13)$$

$$y_j \in \{0, 1\} \quad (3.14)$$

$$b_{jit} \ge 0,\ d_{jit} \ge 0,\ S_t \ge 0,\ u_{it} \ge 0 \quad (3.15)$$

Problem formulation (3.7)-(3.15) is a mixed integer linear programming problem because : except integer variable y_j, all the other variables are continuous.

3.7 The Requirements of Input Data

In this section, we elaborate on data requirements. This will be followed by a numerical example of our model (i.e., section 3.8) and the comparisons of our integrated resource planning type model with the generation expansion planning type model in Ammons and McGinnis [1985] (i.e., section 3.9). There are a large amount of input data requirements for the implementation of our basic model. For currently existing generation units, the inputs required in the model are the following:

-total fixed costs

-upper and lower bounds for the production level of each generation unit

-availability factor for each generation unit

-discounted production cost per unit of energy output for each generation unit

For expansion projects, in addition to the inputs listed above, we need the following extra information:

-construction costs associated with each time period of construction

-time window for initiation of construction

-the length of time needed for construction

For each time period in the whole planning horizon, we need

-number of hours in each interval of each period

-construction budget

-short term rate of return

-inflation rate

-demand forecast

-the demand-side management cost for one unit energy saved in each interval of each period

-utility outage cost in each interval of each period

For the whole model, we need

-number of years in planning horizon

-reserve margin

-the coefficients of the multiple regression model for the expected amount of forced outage

Compared to the input requirements of the model in Ammons and McGinnis [1985], we note that the key differences are that demand-side management, outage cost information, and the regression model for the expected amount of forced outage are required in our model. On the other hand, the model in Ammons and McGinnis [1985] needs loss of load probability criterion. Because we consider forced outage cost as a component in the objective function of a cost minimization model, we do not have loss of load probability criterion.

3.8 Numerical Example

The model is examined in two phases. In this section, we present a preliminary integrated resource planning problem with two currently existing generation units, two projects, and one year planning horizon. In Chapter 5, we solve an integrated resource planning problem with more projects and a longer planning horizon. When the size of the integrated planning problem becomes big, there may be an enormous number of projects that need to be evaluated over a long period of time and thus the computation for the optimal solution easily becomes overwhelming. Therefore, an appropriate optimization algorithm and corresponding computer software package are needed (see., Chapter 5).

Now, we test this model by a simple numerical example. First, we need to estimate the coefficients of the multiple regression model for the expected amount of forced outage. We will use the three data points of Table 3.1 from Hobbs's paper [1995b] (i.e., demand and capacity). By using the way of calculating the probability of getting the expected amount of capacity that is introduced in Ammons [1981], we can get the expected amount of outage. This implies that the data formatted as above can be obtained from published sources. We note that, except the first three data points, the rest of the data of Table 3.1 are hypothetical (i.e., we wanted to use more than three data points for our regression).

As mentioned before, the expected amount of forced outage can be expressed as

$$FO_{it} = \alpha \sum_{j=1}^{J} a_{jt} U_{jt} y_j + \beta P_{it} + \gamma$$

By using SAS programming, we estimate coefficients as $\alpha = -0.009502$, $\beta = 0.044012$, and $\gamma = 0.459070$. We assume that the planning horizon of this numerical example is one year

Table 3.1 Outage Data

Amount of Outage (mw)	Expected Demand (mw)	Expected Capacity (mw)
3.1585	126	200
3.62	81	125
2.832405	89	100
3.33	100	150
1.2245	106	150
2.779	69	75
4.75	95	150
4.0875	115	150
0.39	13	25
1	20	50
6.175	115	150

without time intervals. Moreover, in this generation system, there are only two currently existing generation units and two projects. Each project is to construct a new generation unit. For simplicity, we assume that all selected projects for investment will start at year zero. Because the planning horizon is one year, we do not need to consider the commissioning time for the construction of new generation units. Therefore, these two projects represent two potential new generation units. Table 3.2 lists the numerical data for existing generation units. These data are mainly taken from the book of Electric Plant Cost and Power Production Expenses 1990. The costs listed in these tables are converted to present value. Moreover, because the availability factor in the model only refers to forced outage, we have a forced outage rate for each existing generation units in Table 3.2. Mathematically, forced outage rate is equal to (1 - availability factor).

Table 3.3 lists hypothetical data for projects.

The data for the discounted production cost per unit of energy output are based on Hobbs and Centolella [1995] and are listed in Table 3.4.

In addition, the hypothetical data for construction budget, customer demand, rate of return, and reserve margin are listed in Table 3.5.

We can solve this simple problem through an exhaustive enumeration method. First, we fix the zero-one integer variables . In this example, y_1 and y_2 are fixed at one because they are

Table 3.2 Existing Generation Units

Sources	Fixed Costs US$	Capacity mw	Lower Bound mw	Number of Hours	Forced Outage Rate
Unit 1	57356400	1073	500	8760	0.15
Unit 2	25000000	598	200	8760	0.24

Table 3.3 Potential Generation Units

Sources	Fixed Costs US$	Capacity mw	Lower Bound mw	Number of Hours	Forced Outage Rate	Construction Costs US$
Project 1	50000000	900	450	8760	0.05	100000000
Project 2	30000000	660	300	8760	0.05	150000000

existing generation units. So, the only unknown zero-one integer variables are y_3 and y_4, this results in four possible cases. Then, we solve linear programming problem for each case and the case with the minimum cost will become the final capital budgeting decision.

We assume that the demand-side management cost SO_{it} for one unit energy saved is \$0.065/kwh (i.e., a similar amount of demand-side management cost can be seen in Hobbs and Centolella [1995c]). Moreover, forced outage cost per kwh to a utility is equivalent to the averaged price of electricity over all customer types of a utility. We assume that forced outage cost to a utility is \$1/kwh.

By using MATLAB (i.e., Grace [1992]), we solve all four possible cases. The case with the construction of potential generation unit 1 (project 1) has the minimum total cost. Thus, we claim that our best capital budgeting decision is to construct potential generation unit 1 only. The details of the solution are listed in Table 3.6 (i.e., because the planning horizon is one year without time intervals, we ignore the index i and t.).

Table 3.4 Production Cost Per Unit of Energy Output

Sources	Production Cost \$/mwh
Unit 1 (j=1)	$F_1 = 76.4$
Unit 2 (j=2)	$F_2 = 76.4$
Project 1 (j=3)	$F_3 = 90$
Project 2 (j=4)	$F_4 = 90$

Table 3.5 Miscellaneous Data

Parameter	Value
Construction budget,B_t	\$300000000
Expected demand,P_{it}	2000 mw
Short term rate of return,r_t	10%
Reserve margin,m	5%

3.9 Numerical Example of the model in Ammons and McGinnis

Because the basic model formulated in this chapter is based on the generation expansion planning model developed by Ammons and McGinnis [1985], comparisons and contrasts between the numerical example of our model and the model in Ammons and McGinnis [1985] are presented in this section.

Because of the quadratic production function form, we need extra data for the coefficients of quadratic production functions. The data are based on Li and David [1994] and are listed in Table 3.7.

Then, by using the same data and procedure explained in Section 3.8, we solve Ammons and McGinnis's generation expansion programming problem. The results are listed in Table 3.8 (i.e., similar to Table 3.6, we ignore the index i and t).

By comparing the results of these two models, we find that the total cost of our model is less than the total cost of Ammons and McGinnis [1985]. In detail, both fixed cost and production cost of our model are less than those of the model in Ammons and McGinnis [1985]. That is because we generate less power than the model in Ammons and McGinnis [1985]. In our model, we generate 1150 mw of power (i.e., this amount of power is generated by one new generation unit and two existing generation units), the other 850 mw of customer demand is contributed by demand-side management. On the other hand, in the model in Ammons and McGinnis [1985], two new generation units need to be constructed because all 2000 mw of the customer demand must be satisfied by utility itself. More specifically, based on the capacity of each new generation unit, in the model in Ammons and McGinnis [1985], only the construction of two new generation units plus two existing generation units can provide enough power to

Table 3.6 Optimal Solution of the Basic Model

Item	Value
y_1	1
y_2	1
y_3	1
y_4	0
x_1	500 mw
x_2	200 mw
x_3	450 mw
b_1	500 mw
b_2	200 mw
b_3	450 mw
d_1	0
d_2	0
d_3	0
u	850 mw
S	\$200000000
Fixed cost	\$132356400
Production cost	\$823264800
Demand side management cost	\$483990000
Forced outage cost	\$590197045
Total cost	\$2029808245

satisfy the customer demand.

If the cost of implementing demand-side management for saving a specific amount of the customer demand is greater than the production cost for producing the same amount of power, it is possible that we may get a smaller total cost from Ammons and McGinnis than from our basic model. However, to be consistent with the popularity of demand-side management in the utility industry, we verify the necessity of an integrated resource planning type model (i.e., a model with demand-side and supply-side management).

Table 3.7 Coefficient Information

Sources	v_{1j}	v_{2j}	v_{3j}
Unit 1 (j=1)	0.5	0.05	0
Unit 2 (j=2)	0.75	0.075	0
Project 1 (j=3)	0.5	0.05	0
Project 2 (j=4)	0.75	0.075	0

Table 3.8 Optimal Solution of the Model in Ammons and McGinnis

Item	Value
y_1	1
y_2	1
y_3	1
y_4	1
x_1	600 mw
x_2	400 mw
x_3	600 mw
x_4	400 mw
S	\$50000000
Fixed cost	\$162356400
Production cost	\$5257051724
Total cost	\$5419408124

Via the optimal solution shown in Table 3.8, we see that the total cost is composed of fixed and production costs. However, that does not mean that we do not have forced outage in the model in Ammons and McGinnis [1985]. As stated in Section 3.5, even though we have a strict, high reliability level, forced outage still may occur. The reliability constraints (i.e., reserve margin and loss of load probability) in the model in Ammons and McGinnis [1985] will increase the reliability of power supply of utilities, but they cannot keep utilities free from the occurrence of forced outage. Telson [1975] says, "without regard to how large one chooses to make the system reserve margin, there is always a probability that demand may not be met." Hence, the total cost in the model in Ammons and McGinnis [1985] cannot reflect the real cost of utilities. So, in addition to the consideration of demand-side management, we suggest to have expected outage cost as a component of the objective function to more accurately reflect the real situation of the utility industry.

4 QUANTIFIED OUTAGE COSTS

4.1 Introduction

The capital budgeting process for electric power utilities has been extensively studied. The papers found in the literature, however, primarily focus on relatively easy-to-quantify factors such as construction, operation, and maintenance costs. At the same time, the quantification of the costs associated with outage (a relatively hard-to-quantify factor) has also been extensively studied because of the substantial financial impact of outage on the utilities and customers. In this chapter, we examine the approaches of quantifying the cost of outage. Next we examine the critical issues concerning the use of quantified outage costs (e.g., aggregation of outage costs from different approaches). Then we show how a single approach can be applied for all customer types (i.e., residential, industrial, and commercial).

4.2 Literature Review of Outage Cost Quantification Studies

The literature on the costs of electricity outages can be dated back to 1960s. In response to the important role of outage costs in utility operations and planning, many approaches have been developed for estimating outage costs up to present. Woo and Pupp [1992] found that the outage cost estimates are diverse due to dissimilar measurement concepts, outage attributes and customer demographics in the databases used in various studies. However, the approaches found in the literature can be categorized into three groups: (1) proxy methods, (2) sophisticated econometric methods, and (3) survey methods. This classification is based on the nature of how outage costs are estimated. Proxy methods use secondary data to estimate outage costs. The examples of proxies in the literature include backup power costs and wage rates. Sophisticated econometric methods examine the possible loss to firms in case of outage.

For instance, when an outage occurs the possible outage costs to a firm could include foregone profit, damage to material, and wage payment to idle workers. Survey methods, on the other hand, ask customers to identify their responses in different hypothetical scenarios. Finally, we note that the study of outage cost has often focused on a particular customer type such as residential, industrial, and commercial types.

4.2.1 Residential type

Several papers have argued that the main component of outage costs to residential customers is the loss of leisure. Munasinghe [1980] demonstrates that the enjoyment of leisure is usually constrained in the evening. Leisure activities such as television watching and reading need electricity. Hence, there will be very few substitution possibilities for those electricity-dependent activities if an outage occurs. Munasinghe [1980] also develops a model of estimating the outage costs to the residential customer type by using a utility function. The idea is that an outage disrupts the preferred pattern of consumption and thus leads to a loss. He claims that wage rate could be a good estimate for residential outage costs.

4.2.2 Industrial type

Bental and Ravid [1982] claim,"industrial customers would purchase the backup generating power if the expected gain from the marginal self-generated kwh equals the expected loss from the marginal kwh that is not supplied by the utility. Hence, the expected marginal cost of backup power may serve as an estimate for the marginal outage costs" (i.e., a proxy method). Another commonly used approach is the sophisticated econometric approach. Tishler [1993] develops a model with a production function to measure expected outage costs. In that paper, various sources such as a possible reduction in productivity contribute to the costs of outage.

4.2.3 Commercial type

Sanghvi [1982] finds that several problems may arise in developing a proper method to estimate the outage cost for commercial customers because the definition of commercial activ-

ities is often unclear. Sanghvi claims that "large apartment buildings, small pizza parlors, and moderate size manufacturing firms can be classified as the commercial user types." However, a large portion of commercial activities are production activities. We can estimate the outage costs of those activities by using the same approaches as we do for the industrial customer type. Moreover, some commercial activities, such as large shopping centers, can be regarded as having characteristics similar to the residential customer type and can be analyzed by using the residential approaches.

4.3 Difficulties in Reconciling Different Approaches for Use in Capital Budgeting Process

The diversity of quantification approaches makes it difficult for utilities to evaluate capital investment projects analytically and objectively. If different approaches are employed for different customer types, and if the resulting costs of outage are aggregated, this could present a serious problem to the decision maker of the capital investment. For example, let us suppose that Tishler's model [1993] (a sophisticated econometric method) is used to estimate the outage costs of industrial customers. At the same time, let us suppose that Munasinghe's model [1980] (a proxy method) based on the wage rate is used to estimate the outage costs of residential customers. Finally, let us suppose that a survey method is used for commercial customers. Aggregating the outage costs over all customers in this case will be difficult to justify because (1) underlying assumptions are quite different (e.g., one of Tishler's assumptions on outage is that the length of outage is a random variable with a known distribution function relative to Munasinghe [1980]), and (2) resulting outcomes have their own unique tendencies (e.g., survey methods tend to overestimate; see, e.g., Sanghvi [1982]).

A few papers have demonstrated that we should be careful when we compare outage costs. Tishler [1993] says, "direct comparison of outage cost estimates from different studies should be viewed with caution." Moreover, Sanghvi [1982] states,"there exists the factors that sometimes limit the ability to strictly compare different outage cost estimates. Often the estimates are based upon different methodologies, assumptions, economic and demographic mixes and local

conditions, and for different outage descriptors."

4.4 A Single Quantification Approach for All Customer Types

Caves et al. [1990] note that surveys are the primary source of information on customer outage costs. Survey methods ask customers to identify their probable response in different hypothetical scenarios. The customers would be asked how much it would cost them to adjust to this power outage. However, since most consumers do not have much quantification experience and knowledge, it is nearly impossible for them to accurately estimate the costs under the hypothetical scenarios.

Sophisticated econometric methods have been broadly used for estimating industrial and commercial outage costs. But there will be difficult problems if we use them for residential customers. Caves et al. [1990] claim that residential customer type could be viewed as an industrial customer type if we treat household activities as a production process where electricity is an input and leisure activities are outputs. Then residential outage cost is the market value of those household activities lost because of an outage. The problem is that obtaining a complete listing of household activities and their market values is difficult.

In contrast to the survey approach and the sophisticated econometric approach, the proxy approach is conceptually simple and the data for proxies are readily available. Because of these reasons, the proxy approach can be an effective way for first-order approximation and as the data accumulate, the accuracy of this approach can improve. To demonstrate how the proxy approach is applied to quantification of outage costs of all customer types, we will employ the wage rate as the proxy for the outage cost of residential customers and the backup power cost as the proxy for the outage cost of industrial and commercial customers.

4.4.1 Wage Rate Method for Residential Customers

Munasinghe [1980] demonstrates that the wage rate can be a good estimate for residential outage costs. The mathematical proof for this relation starts from developing the utility function of a household. Because the loss of leisure is viewed as the main component of outage

costs in Munasinghe [1980], the utility of a typical household is expressed as a function of leisure types and income. In his paper, however, two types of leisure are considered. One cannot be enjoyed without electricity, the other is electricity independent. By maximizing the utility function subject to a linear budget constraint, he found that the incremental monetary values of both types of leisure can be roughly estimated by the wage rate. Therefore, he formulates a linear regression model as follows:

$$OC_i = b_1 + b_2 Y_i + U_i$$

where i, OC_i, U_i, and Y_i are the index of customer, the outage cost per hour for customer i, the random disturbance term for customer i, and the net income earning per hour for customer i. Then Munasinghe [1980] selects 27 residential customers as a sample from the city of Cascavel, Brazil, and uses questionnaires to collect the data of outage costs. The collected data are shown in Table 4.1 using Cruzeiros (Brazilian currency) represented by Cr$. In the end of 1976, one U.S. dollar was equal to twelve Cr dollars and thirty five Cr cents.

By using the computer program SAS, we verified that the estimated coefficients are $b_1 = 3.355$ and $b_2 = 0.896$, yielding a good fit with $R^2 = 0.901$. Hence, the proxy relation between the outage cost and the wage rate for customer i is $OC_i = 3.355 + 0.896 Y_i$.

4.4.2 Backup Power Cost for Industrial and Commercial Customers

Thus far we have shown how the outage cost is estimated for the residential customers via the wage rate method. Now we will demonstrate how the outage cost is estimated for the industrial and commercial customers via the backup power cost method. This method is based on work by Bental and Ravid [1982].

Bental and Ravid [1982] say, "The expected gain from the marginal self-generated kwh is also the expected loss from the marginal kwh which is not supplied by the utility. Therefore, the marginal cost of generating its own power may serve as an estimate for the marginal outage cost". The cost of generating backup power consists of two items: the cost of constructing the backup facility, and the operating cost. So the expected marginal cost of a kwh generated can

Table 4.1 Residential Consumer Survey Results

Net Income Earnings Rate(Cr$ per hour)	Outage Cost (Cr$ per hour)	Net Income Earning Rate(Cr$ per hour)	Outage Cost (Cr$ per hour)
7.4	6	27.7	20
7.4	7	30.8	35
7.4	13	40.3	30
10.5	12	40.3	30
11.1	15	45.5	50
14.2	10	48.0	60
15.4	20	55.1	45
18.5	25	64.6	55
18.5	35	64.6	60
19.7	20	73.9	85
21.5	15	99.1	60
21.5	25	116.3	133
24.6	15	166.2	150
24.6	40		

be expressed as

$$MC = \frac{b'(Kg)}{H} + v$$

where $b'(Kg)$, H, and v are the annualized cost of backup facility per kw, the expected duration of outage per year, and the operating cost of generating power per kwh. However, we illustrate the following example to help the reader understand the concept of backup power cost method. This example is taken from Bental and Ravid's paper [1982]:

Example:

The cost of the backup facility for one kw of backup power is \$106. At straight-line depreciation over ten years, a tax rate of 50%, and an interest rate of 8%, the yearly cost per generator KW is \$10.50 $[b'(Kg)]$. The expected total duration of outages, H, was taken to be ten hours per year. Operating costs are \$0.11 per kwh (v). Using the above equation, we obtain a figure of about \$1.16 per unsupplied kwh.

Bental and Ravid [1982] present the sensitivity of the estimated marginal cost of an unsupplied kwh to different assumptions on H and r, where $b(Kg) = \$106$ per kw and $v = \$0.10$ per kwh, depreciation is taken linearly over ten years, and the tax rate is 50%. The resulting

Table 4.2 Marginal Outage Costs

H	$r = 0.08$	$r = 0.10$	$r = 0.12$	$r = 0.20$
5	2.20	2.49	2.79	4.10
10	1.15	1.29	1.45	2.10
20	0.62	0.70	0.77	1.10
40	0.36	0.40	0.44	0.60
100	0.20	0.22	0.24	0.30

marginal outage costs (represented by the backup facility cost per kwh plus the operating cost of generating power per kwh) are shown in Table 4.2.

4.5 The Use of Quantified Outage Costs in Integrated Resource Planning Models

We now proceed to explain how the quantified outage cost to a utility is used in an integrated resource planning model. As stated in Section 3.3, the expected amount of forced outage is determined by the levels of customer demand and the available capacity of a utility. If we multiply the expected amount of forced outage by the outage cost per kwh to a utility and the number of hours for the operation of a utility in a time interval, we can obtain the utility forced outage costs. On the other hand, by using the quantification approach introduced in Section 4.4, we can find the customer outage cost for one unit power outage. Similarly, if we multiply the expected amount of forced outage by the outage cost per kwh to customers and the number of hours for the operation of a utility, we can find the customer forced outage cost. However, the objective function in our basic model (see., Chapter 3) is to minimize the total cost to a utility. In other words, only utility outage cost is considered in our model. For future extensions, it is possible to include customer outage cost in our integrated resource planning model (see., Hobbs [1995a]).

4.6 The Treatments for Stochastic Forced Outage in Our Model

As stated in Chapter 3, the expected amount of forced outage can be expressed as

$$FO_{it} = \alpha \sum_{j=1}^{J} a_{jt} U_{jt} y_j + \beta P_{it} + \gamma$$

where FO_{it} values are used as parameters in (3.7). In this multiple regression model, there are two independent variables. One is the expected totally available capacity. The other is the expected customer demand. When both of the independent variables are equal to zero, the expected amount of forced outage is equal to the intercept, γ. If we take the partial derivative of FO_{it} with respect to $\sum_{j=1}^{J} a_{jt} U_{jt} y_j$, we can get α. If α is less than zero, then we say that for every unit decrease in the expected totally available capacity, the expected amount of forced outage increases by α. Similarly, if we take the partial derivative of FO_{it} with respect to P_{it}, we can get β. If β is greater than zero, then we say that for every unit increase in the expected customer demand, the expected amount of forced outage increases by β.

For the theoretical interest, we discussed the relationship of reliability and the occurrence of forced outage in Section 3.5. In our discussion, we assumed that both peak demand plus a certain amount of reserve margin as well as available capacity are stochastic (i.e., peak demand can be thought of as the maximum amount of the expected demands among all time intervals in a same period). Also, for simplicity, we assumed that they are uniformly distributed (i.e., Ammons [1981] assumed that available capacity and peak demand can be approximated by Normal distributions). Then we found that there exists a probability of having forced outage while the constraints (3.3) (i.e., reliability constraints) are satisfied. Thus, we considered forced outage cost as a component of the utility's total cost in our basic model.

For the ease of computation, we treated the available capacity and customer demand as deterministic quantities in our basic model in Chapter 3 (i.e., the similar deterministic treatment for the inclusion of random outages can be seen in the mathematical model in Hobbs [1995a]). We defined $\sum_{j=1}^{J} a_{jt} U_{jt} y_j$ as the expected totally available capacity in period t and P_{it} as the expected customer demand prior to the implementation of demand-side management in interval i of period t. In other words, the mean expected quantities of the two random

variables are used in our basic model (i.e., P_{it} and a_{jt} are stochastic).

For the numerical example of our basic model stated in Section 3.8, we can get the information of totally available capacity, customer demand, and the corresponding customer demand from the past performance record of utilities. Then we can use them as given numbers for the expected totally available capacity and the expected customer demand of our basic model. Furthermore, we can estimate α, β, and γ of our multiple regression model (econometric model) by using statistics software package. Here, it is important to note that, for the implementation of our econometric model, the independent variables are given numbers, and not random variables. However, this is an attempt to treat the random features of forced outage in a deterministic way.

5 COMPUTATIONAL ALGORITHM AND COMPARISON TO OTHER MODELS

5.1 Introduction

We used the exhaustive enumeration method to find the optimal solution for the mixed integer linear programming problem in Chapter 3. However, for a large-scale integrated resource planning problem, the number of possible combinations of projects might be very large and difficult to be solved by the exhaustive enumeration method. Therefore, in this chapter we will introduce the branch-and-bound algorithm, and a relevant computer software package. Then we will solve two numerical examples for our basic model. The comparisons and contrasts of the results of these examples are provided in Appendix B.

5.2 Branch-and-Bound Algorithm

Branch-and-bound algorithm is one of the most commonly used algorithms for global optimization. It has been developed through the efforts of many researchers since the 1960's. The branch-and-bound technique has been broadly used for solving integer programming problems such as binary integer programming, pure integer programming, and mixed integer linear/nonlinear programming problems. Levary and Seitz [1990] say, "the branch-and-bound method for solving integer programming problem is perhaps the most reliable". Because our basic model introduced in Chapter 3 is a mixed integer linear programming problem, we choose the branch-and-bound method to solve our basic model. Based on the materials of Gupta and Ravindran [1985] and Hillier and Lieberman [1990], we provide an overview of the branch-and-bound method in the rest of this section. It is important to note that our discussion is for

minimization problems only.

The branch-and-bound algorithm starts by relaxing the integrality requirements of a mixed integer linear programming problem. For a minimization problem, the optimal solution we get from the relaxation of the integrality requirements will become the lower bound on the optimal value of the mixed integer linear programming problem. The concept here is that we want to know how good each problem's best feasible solution can be by solving the relaxation of the problem. However, if our solution for the relaxed problem satisfies all the integrality requirements, then it is the optimal solution to the mixed integer linear programming problem. Otherwise, we should select one of the integer variables that violate the integrality requirements to form two new subproblems.

The process of forming two subproblems is called branching. The variable used to do branching is called the branching variable. For example, if we select integer variable y_j that appears to be continuous in the feasible solution for the relaxed problem as the branching variable, we divide the value of y_j into integral $[y_j]$ and fractional parts $\bar{y_j}$. Mathematically, $y_j = [y_j] + \bar{y_j}$, where $[y_j]$ is an integer and $0 < \bar{y_j} < 1$. From here, we form two subproblems. One is the original problem plus the additional constraint $y_j \leq [y_j]$. The other is the original problem plus the additional constraint $y_j \geq [y_j]+1$. We call this process of forming subproblems as branching.

After we branch the original problem into two subproblems, similarly, we relax the integrality requirements of these two subproblems and solve these two continuous problems. If the relaxed subproblem has a feasible integer solution, then the corresponding value of the objective function becomes an upper bound of the mixed integer linear programming problem. We call this solution as an incumbent (i.e., Hiller and Lieberman [1990] define incumbent as "the best feasible solution found so far"). If the value of the objective function of a relaxed subproblem is greater than or equal to the current incumbent, then this subproblem should be dismissed from further consideration. This act is called fathoming. The reason for doing this is that we should not consider a subproblem whose lower bound for the feasible integer solution of the subproblem is greater than or equal to the value of the objective of a currently

existing feasible integer solution for the same minimization problem.

Hiller and Lieberman [1990] say, "a subproblem can be conquered (fathomed), and thereby dismissed from further consideration." In addition to the comparisons with the current incumbent, there are two other ways of fathoming. One is if the relaxed subproblem has no feasible solutions. The other is if the relaxed subproblem has a feasible integer solution. In this case, if the solution of this relaxed subproblem is better than the current incumbent, then it becomes a new incumbent (a new upper bound).

Generally, the solutions for the original problem and subproblems are stored in the corresponding nodes in the tree of showing the branch-and-bound process. A node stores the information of the bound and the optimal solution for the corresponding relaxed subproblem. However, the search for the optimal solution of a mixed integer programming problem terminates when all the nodes are fathomed. We have a branch-and-bound tree in Figure 5.1.

The general framework of the branch-and-bound algorithm can be applied to all types of integer programming problems. However, wherever it is used, the branch-and-bound algorithm has the same three basic steps of branching, bounding, and fathoming. The difference is how these steps are performed.

5.3 Branch-and-Bound Algorithm for Capital Budgeting Models

The basic model in Chapter 3 was formulated for capital budgeting problems. The investment projects in the model are either accepted or rejected. Therefore, the decision variables for project selection are restricted to zero-one. In order to solve this capital budgeting type of mixed integer linear programming problems (i.e., mixed binary integer linear programming problem), some slight changes of the branch-and-bound algorithm introduced in Section 5.1 are needed.

As stated before, the branch-and-bound algorithm starts by relaxing the integrality requirements of a mixed integer linear programming problem. However, in the beginning a mixed binary integer linear programming problem should be solved as a linear programming problem by replacing the zero-one constraints by continuous constraints. For example, constraints

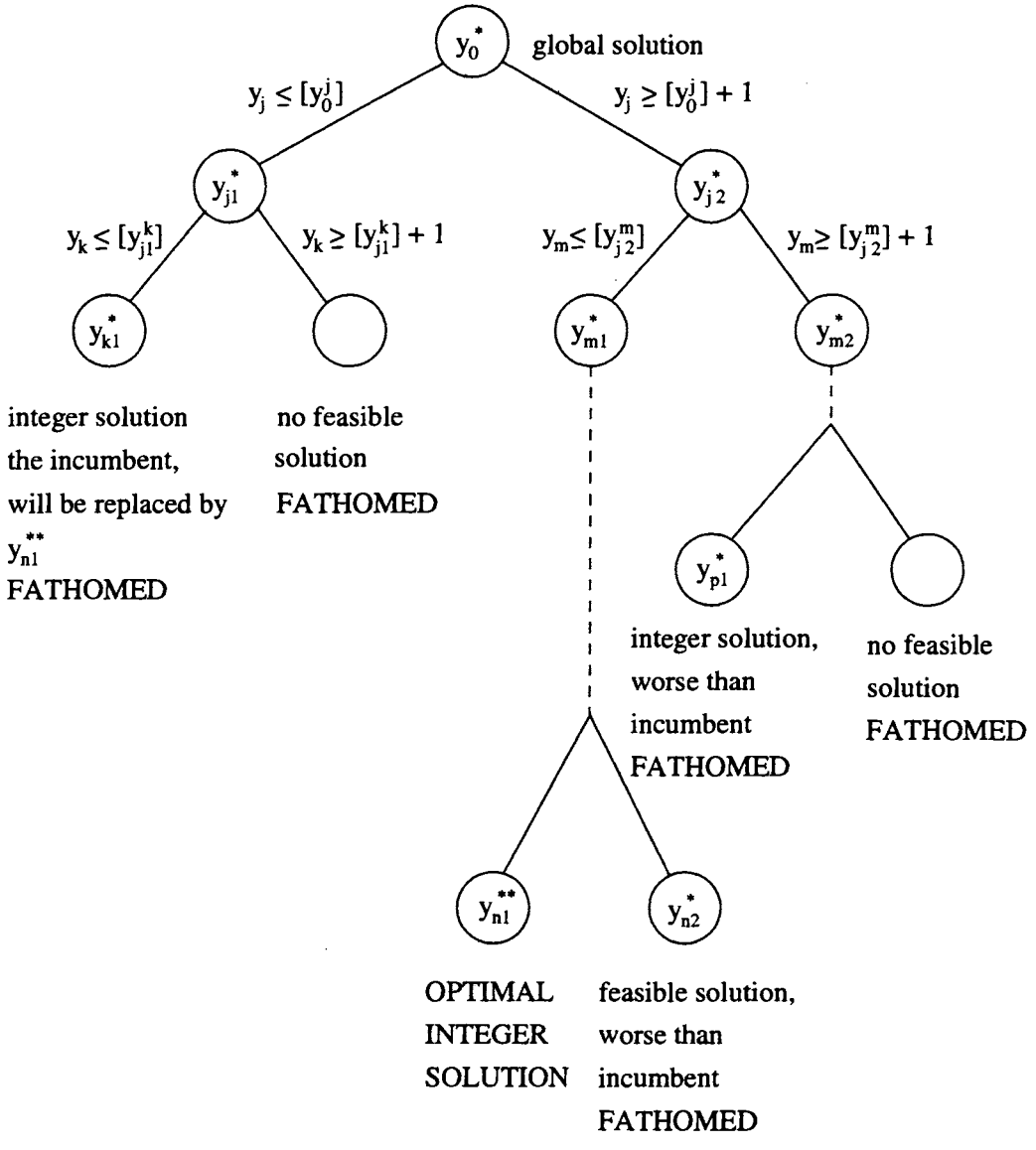


Figure 5.1 The Branch-and-Bound Algorithm

$y_j \in \{0, 1\}$ in Section 3.6 should be replaced by the following constraints: $0 \leq y_j \leq 1$. If the solution of this problem includes fractional investment projects, then we choose one as the branching variable and form two subproblems . For example, if we choose y_j as the branching variable, then one subproblem is the original problem plus the additional constraint $y_j = 0$. The other subproblem is the original problem plus the additional constraint $y_j = 1$. By using the principles of the branch-and-bound method shown in Section 5.1, we can solve the capital budgeting type of mixed integer linear programming problems.

5.4 Software for Mixed Integer Linear Programming Problems (MILP)

A recent optimization software, LINGO, has been proved as an efficient tool to solve linear optimization models with binary decision variables (see, Sarper [1995]). In other words, LINGO might be an appropriate tool for solving our capital budgeting problems.

According to the LINGO user's guide [1995], we know that LINGO is designed to solve integer programming problems by using the branch-and-bound algorithm. In Chapter 3, we solved the numerical example by using the exhaustive enumeration method. After we fixed the integer variables, we solved each individual linear programming problem by MATLAB. However, by using the same numerical example as in Chapter 3, LINGO solves the problem in one second of runtime and has an identical solution. However, for nonlinear programming problems, LINGO is not as good for linear programming problems. In Chapter 10 of the LINGO user's guide [1995], it says, "when LINGO finds a solution to a linear optimization model, it is the definitive best solution: we say it is the global optimum. This is not the case for nonlinear optimization." Also, it says, "For nonlinear problems, a solver can only identify a local optimum. Therefore, in nonlinear optimization models it is important to understand that the solution returned by a solver may be locally optimal, but may not be the global optimal solution."

5.5 The Numerical Examples for MILP Integrated Resource Planning

By using the LINGO software package (i.e., the branch-and-bound algorithm), we test two numerical examples in this section. One problem has five time periods with no time intervals as the planning horizon (i.e., five years), four currently existing generation units and six potential new generation units in the system. In this example, we assumed that all the constructions for new generation units occur at year one. In other words, six potential new generation units are equivalent to six projects. The other problem has five time periods with no time intervals as the planning horizon (i.e., five years), four currently existing generation units and six potential new generation units. Importantly, six potential new generation units can be constructed at any year during the planning horizon. In other words, a particular potential generation unit is associated with five projects. Hence, there are thirty projects in this example.

For the first problem, there are 161 variables including 6 integer variables, and 161 constraints. LINGO solves this example in 102 iterations in 2 seconds. For the second problem, there are 545 variables including 30 integer variables, and 527 constraints. LINGO solves this problem in 532 iterations in 12 seconds. The details of these two numerical examples are provided in Appendix A and Appendix B.

5.6 Comparison to Other Models

We list the comparisons and contrasts among the model in Ammons and McGinnis [1985], the model in Hobbs [1995a], and our model in Table 5.1.

In Table 5.1, MINLP stands for mixed integer nonlinear programming. MILP stands for mixed integer linear programming. LOLP stands for loss of load probability. Finally, A&M stands for Ammons and McGinnis.

An obvious difference among the three models is how they attempt to satisfy customer demand. The model in Ammons and McGinnis [1985] attempts to satisfy customer demand by constructing new generation units. On the other hand, the model in Hobbs [1995a] and our model attempt to satisfy customer demand from both demand-side and supply-side management. In short, one is a traditional supply-side management model. The other is an integrated

Table 5.1 Comparison to Other Models

Features	**A&M**	**Hobbs**	**Our Model**
Minimize Total Cost	to utility	to utility and its customers	to utility
Planning Strategy	supply-side	demand-side and supply-side	demand-side and supply-side
Problem Type	MINLP	MILP	MILP
Reliability Criteria	LOLP and reserve margin	reserve margin	reserve margin
Construction Budget	across period	none	across period
Production Cost Representation	quadratic	constant	constant
Inclusion of Power Surplus	yes	no	yes
Inclusion of Power Deficit	no	yes	yes
Inclusion of Forced Outage Cost	no	no	yes
Inclusion of Scheduled Outage Cost	no	yes	no

resource planning type model.

Even though both our model and the model in Hobbs [1995a] are integrated resource planning type models, there still exists some critical difference. One is that Hobbs has scheduled outage cost as the cost to the customers of a utility (c.f., forced outage cost is a component of the total cost to a utility in our model). The other is that Hobbs always assumes the existence of power deficit (c.f., by dividing production level to power generated and delivered, and power generated but not delivered, both power surplus and power deficit are possible in our model).

6 CONCLUSIONS

6.1 Summary

With the complexity and the rapid change of the utility industry, a few issues need to be considered when utilities make plans for satisfying customer demand. In Chapter 1, we briefly introduced the issues that are related to our model formulation. In Chapter 2, we reviewed the relevant literatures including generation expansion planning, reliability, and integrated resource planning. Based on the comparisons and contrasts of a generation expansion planning model and an integrated resource planning model, we formulated our basic model in Chapter 3. Because outage costs are included in our model, in Chapter 4 we presented different approaches for quantifying outage costs. In Chapter 5, we introduced the branch-and-bound algorithm for solving our mixed integer linear programming problem and a solvable software package.

6.2 Contributions

In this thesis, a comprehensive integrated resource planning model combining demand-side and supply-side management was developed. We compared and analyzed two types of concepts of designing mathematical models for meeting the ever-increasing customer demand (Chapter 2). One type is the generation expansion planning which is based on supply-side management. The other type is the integrated resource planning which includes demand-side management in the framework of supply-side management. Then we developed a comprehensive integrated resource planning model with the features of both types of models (Chapter 3). Different from other integrated resource planning models, our model considered forced outage cost as a component of the total cost to a utility. By using the stochastic features of forced outage, we claimed that the coexistence of both reliability constraints and forced outage in our model is

reasonable. Moreover, we considered production level of utilities as the sum of power generated and delivered to customers and power generated but not delivered to customers. This concern is consistent with the features of power deficit and power surplus. Therefore, it was demonstrated that our integrated resource planning model considers the economic activities of the present utility environment and can reflect the real cost of a utility.

In Chapter 4, we discussed the quantification of outage costs. Outage costs have been thought as a hard-to-quantify factor. We discussed the quantification approaches for estimating the outage costs of different customer types. Then we suggested a quantification approach which can be used for all customer types. However, because the objective of our integrated resource planning model is to minimize the total cost of a utility, we only considered utility outage cost in our model. However, it is possible that we can extend our model to include customer outage cost.

6.3 Future Research

A number of extensions of our model can be proposed. We present outside purchase and sale as the topic for future research.

With the arrival of the era of deregulation, outside purchase and outside sale for power have become popular economic activities among utilities. According to the definitions of power deficit and power surplus in section 3.4, we note that power purchase is a way of eliminating the power deficit of utilities. Similarly, power sale is a way of dealing with the power surplus of utilities.

We now have two new decision variables to add to our basic model. One is the amount of the power through outside purchase. The other is the amount of power through outside sale. Because now generation units (i.e., existing and new generation units) are no longer the only sources of power, we should consider outside purchase and outside sale as contributors to the available capacity of utilities as well. In this case, however, it is inevitable that we must consider transmission systems.

In this thesis, we only discuss generation systems. However, transmission systems also

have similar features as generation systems. For example, in a transmission system, utilities also need to meet customer demand. The demand satisfaction constraints become complicated in transmission systems because we need to consider power importing regions and power exporting regions. Similar to generation systems, there are fixed cost and production cost in transmission systems. However, the separate studies of generation and transmission systems do not necessarily provide an optimal solution for the combination of generation and transmission systems.

APPENDIX A NUMERICAL EXAMPLE 1

We have two numerical examples in Chapter 5 for the implementation of our basic model. It is important to know that we ignore the construction time for potential new generation units. In other words, we assume that a potential generation unit starts to produce power from the commissioning year of its construction. However, there are no difficulties with considering positive periods for construction in our model. As stated in Section 3.4, for the implementation of our model, we should fix lots of corresponding parameters at zero for a specific potential generation unit before the start of its construction. Hence, in the case of considering construction time, what we need to do is to fix parameters at zero for additional time periods. These extra time periods are the construction time.

In this appendix, we present the data of the first example and solve for the optimum solution by using LINGO.

Numerical Example 1

As stated in chapter 5, this example has five time periods without time intervals as the planning horizon (i.e., five years), four currently existing generation units and six new potential generation units. Because we assume that all the constructions for new generation units occur at year one, there are only six projects in this example (i.e., $J = 10$, the first four are existing generation units, the last six are potential generation units(projects)).

The data of fixed cost for existing generation units are US$57,356,400, US$25,000,000, US$45,000,000, and US$30,000,000 for generation units 1 (f_1), 2 (f_2), 3 (f_3), and 4 (f_4), respectively. The data of fixed cost for potential generation units are $50,000,000 ($f_5$), $30,000,000 ($f_6$), $25,000,000 ($f_7$), $15,000,000 ($f_8$), $17,000,000 ($f_9$), and $17,000,000 ($f_{10}$). The reserve

margin (i.e., m) is 0.05.

According to the requirements of input data in Section 3.7, other hypothetical data are listed in Tables A.1 through A.9 (because there is no time intervals, we ignore the index i). Importantly, because all the constructions occur at year one, constraints (3.11) and (3.13) in our basic model in Section 3.6 are not considered. By using LINGO, we solve our basic model in section 3.6 for this numerical example. The optimal solution of this example is listed in Tables A.10 through A.17.

Table A.1 Yearly Data

Data	**Year1**($t=1$)	**Year2**($t=2$)	**Year3**($t=3$)	**Year4**($t=4$)	**Year5**($t=5$)
H_t(hour)	8760	8000	7800	8460	8600
SO_t(\$/mwh)	65	80	81	85	90
FOC_t(\$/mwh)	1000	2000	2100	2400	2600
P_t(mw)	2500	3500	4000	5500	6000

Table A.2 Capacity of Existing Generation Units (mw)

Data	**Year1**($t=1$)	**Year2**($t=2$)	**Year3**($t=3$)	**Year4**($t=4$)	**Year5**($t=5$)
U_{1t}	1073	800	800	1200	1200
U_{2t}	598	700	670	670	670
U_{3t}	700	700	700	700	750
U_{4t}	900	900	900	900	950

Table A.3 Availability Factor of Existing Generation Units

Data	**Year1**($t=1$)	**Year2**($t=2$)	**Year3**($t=3$)	**Year4**($t=4$)	**Year5**($t=5$)
a_{1t}	0.85	0.90	0.90	0.90	0.90
a_{2t}	0.76	0.90	0.90	0.90	0.90
a_{3t}	0.85	0.85	0.85	0.85	0.85
a_{4t}	0.85	0.85	0.90	0.90	0.90

From Table A.10, we note that all six potential generation units will be constructed.

Table A.4 Lower Bound on Production Level on Existing Generation Units (mw)

Data	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
L_{1t}	500	350	350	550	550
L_{2t}	200	300	300	300	300
L_{3t}	300	300	300	300	320
L_{4t}	400	400	400	400	450

Table A.5 Production Cost of Existing Generation Units ($/mwh)

Data	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
F_{1t}	86	88	90	92	94
F_{2t}	85	86	87	88	89
F_{3t}	85	86	87	88	89
F_{4t}	90	92	94	96	98

Table A.6 Capacity of Potential Generation Units (mw)

Data	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
U_{5t}	900	900	900	900	900
U_{6t}	660	660	660	660	660
U_{7t}	600	600	600	600	600
U_{8t}	400	400	400	400	400
U_{9t}	500	500	500	500	500
U_{10t}	500	500	500	500	500

Table A.7 Availability Factor of Potential Generation Units

Data	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
a_{5t}	0.95	0.95	0.95	0.95	0.95
a_{6t}	0.95	0.95	0.95	0.95	0.95
a_{7t}	0.95	0.95	0.95	0.95	0.95
a_{8t}	0.95	0.95	0.95	0.95	0.95
a_{9t}	0.95	0.95	0.95	0.95	0.95
a_{10t}	0.95	0.95	0.95	0.95	0.95

Table A.8 Lower Bound on Production Level on Potential Generation Units (mw)

Data	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
L_{5t}	450	400	400	400	400
L_{6t}	350	350	350	350	350
L_{7t}	300	300	300	300	300
L_{8t}	100	100	100	100	100
L_{9t}	200	200	200	200	200
L_{10t}	200	200	200	200	200

Table A.9 Production Cost of Potential Generation Units ($/mwh)

Data	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
F_{5t}	86	87	88	89	90
F_{6t}	90	92	94	96	98
F_{7t}	92	94	96	98	100
F_{8t}	93	94	95	96	97
F_{9t}	94	96	98	100	102
F_{10t}	95	97	99	101	103

Table A.10 Global Optimal Solution and Integer Variables

Items	Value
Objective Value	$29012750000
Integer Variable y_j's	all are one

Table A.11 Demand-side Management Program(mw)

Variable	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
u_t	0	600	1100	2400	2209.5

Table A.12 Production Level of Existing Generation Units (mw)

Variable	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
x_{1t}	500	350	350	550	550
x_{2t}	200	300	300	300	603
x_{3t}	300	300	300	300	637.5
x_{4t}	400	400	400	400	450

Table A.13 Production Level of Potential Generation Units (mw)

Variable	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
x_{5t}	450	400	400	400	400
x_{6t}	350	350	350	350	350
x_{7t}	300	300	300	300	300
x_{8t}	100	100	100	100	100
x_{9t}	200	200	200	200	200
x_{10t}	200	200	200	200	200

Table A.14 Power Generated and Delivered by Existing Generation Units (mw)

Variable	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
b_{1t}	500	350	350	550	550
b_{2t}	0	300	300	300	603
b_{3t}	300	300	300	300	637.5
b_{4t}	400	400	400	400	450

Table A.15 Power Generated and Delivered by Potential Generation Units (mw)

Variable	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
b_{5t}	450	400	400	400	400
b_{6t}	350	350	350	350	350
b_{7t}	300	300	300	300	300
b_{8t}	0	100	100	100	100
b_{9t}	200	200	200	200	200
b_{10t}	0	200	200	200	200

Table A.16 Power Generated but Not Delivered by Existing Generation Units (mw)

Variable	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
d_{1t}	0	0	0	0	0
d_{2t}	200	0	0	0	0
d_{3t}	0	0	0	0	0
d_{4t}	0	0	0	0	0

Table A.17 Power Generated but Not Delivered by Potential Generation Units (mw)

Variable	**Year1**($t = 1$)	**Year2**($t = 2$)	**Year3**($t = 3$)	**Year4**($t = 4$)	**Year5**($t = 5$)
d_{5t}	0	0	0	0	0
d_{6t}	0	0	0	0	0
d_{7t}	0	0	0	0	0
d_{8t}	100	0	0	0	0
d_{9t}	0	0	0	0	0
d_{10t}	200	0	0	0	0

APPENDIX B NUMERICAL EXAMPLE 2

In this appendix, we present the data of the second example and solve for the optimum solution by using LINGO.

Numerical Example 2

This example has five time periods without time intervals as the planning horizon (i.e., five years), four currently existing generation units and six potential generation units. Importantly, all potential generation units can be constructed at any year during the planning horizon. Therefore, there are thirty projects in this example (i.e., $J = 34$, the first four are existing generation units, the last thirty are projects). For example, in this example, potential generation unit 1 is associated with projects 5 through 9. Project 7 is the project of constructing potential generation unit 1 at year three. Similarly, project 34 is the project of constructing potential generation unit 6 at year five.

The data of fixed cost for existing generation units are \$57,356,400, \$25,000,000, \$45,000,000, and \$30,000,000 for generation units 1 (f_1), 2 (f_2), 3 (f_3), and 4 (f_4), respectively. Reserve margin (i.e., m) is 0.05. The data of capacity, availability factor, lower bound on production level, and production cost for existing generation units are listed in Tables A.2, A.3, A.4, and A.5, respectively. According to the requirements of input data in Section 3.7, other hypothetical data are listed in Tables B.1 through B.7 (because there are no time intervals, we ignore the index i). By using LINGO, we solve our basic model in section 3.6 for this numerical example. The optimal solution of this example is listed in Tables B.8 through B.16.

From the optimal solution in Table B.8, we note that y_1, y_2, y_3, and y_4 are equal to one because they are currently existing generation units. The value of one of y_6 means that

Table B.1 Yearly Data

Data	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
H_t(hour)	8760	8000	7800	8460	8600
SO_t(\$/mwh)	65	80	81	85	90
FOC_t(\$/mwh)	1000	2000	2100	2400	2600
P_t(mw)	2500	3500	4000	5500	6000
B_t(US\$10000)	100000	100000	100000	100000	100000
r_t	0.1	0.08	0.08	0.08	0.1

Table B.2 Fixed Cost of Potential Generation Units (US\$)

Items	Fixed Cost	Items	Fixed Cost
f_5	50000000	f_{20}	15000000
f_6	40000000	f_{21}	12000000
f_7	30000000	f_{22}	9000000
f_8	20000000	f_{23}	6000000
f_9	10000000	f_{24}	3000000
f_{10}	30000000	f_{25}	17000000
f_{11}	25000000	f_{26}	14000000
f_{12}	20000000	f_{27}	11000000
f_{13}	15000000	f_{28}	8000000
f_{14}	10000000	f_{29}	5000000
f_{15}	25000000	f_{30}	17000000
f_{16}	20000000	f_{31}	14000000
f_{17}	15000000	f_{32}	11000000
f_{18}	10000000	f_{33}	8000000
f_{19}	5000000	f_{34}	5000000

potential generation unit 1 will be constructed at year two. The value of one of y_{11} means that potential generation unit 2 will be constructed at year two. The value of one of y_{16} means that potential generation unit 3 will be constructed at year two. The value of one of y_{20} means that potential generation unit 4 will be constructed at year one. The value of one of y_{26} means that potential generation unit 5 will be constructed at year two. Finally, the value of one of y_{31} means that potential generation unit 6 will be constructed at year two.

Comparisons and Analyses of Two Numerical Examples

By using a similar data set, we obtained different solutions for two numerical examples. In numerical example 1, we assumed that all the construction for potential generation units occurs at year one. Subsequently, the solutions of binary variables imply that we need to construct all six potential generation units at year one. These results partially caused the value of zero for the amount of power saved by demand-side management programs at year one (see., Table A.11).

In contrast, the solutions of the binary variables in numerical example 2 also imply the construction for all six potential generation units, but they are constructed in different years over the planning horizon. That is probably why the total cost to a utility in example 2 is less than the total cost in example 1, because in numerical example 2 the utility can choose the best timing for constructing new generation units to have the least-cost investments. However, numerical example 2 is the full application of our basic model in Chapter 3.

Table B.3 Capacity of Potential Generation Units (mw)

Data	**Year1**($t=1$)	**Year2**($t=2$)	**Year3**($t=3$)	**Year4**($t=4$)	**Year5**($t=5$)
U_{5t}	900	900	900	900	900
U_{6t}	0	900	900	900	900
U_{7t}	0	0	900	900	900
U_{8t}	0	0	0	900	900
U_{9t}	0	0	0	0	900
U_{10t}	660	660	660	660	660
U_{11t}	0	660	660	660	660
U_{12t}	0	0	660	660	660
U_{13t}	0	0	0	660	660
U_{14t}	0	0	0	0	660
U_{15t}	600	600	600	600	600
U_{16t}	0	600	600	600	600
U_{17t}	0	0	600	600	600
U_{18t}	0	0	0	600	600
U_{19t}	0	0	0	0	600
U_{20t}	400	400	400	400	400
U_{21t}	0	400	400	400	400
U_{22t}	0	0	400	400	400
U_{23t}	0	0	0	400	400
U_{24t}	0	0	0	0	400
U_{25t}	500	500	500	500	500
U_{26t}	0	500	500	500	500
U_{27t}	0	0	500	500	500
U_{28t}	0	0	0	500	500
U_{29t}	0	0	0	0	500
U_{30t}	500	500	500	500	500
U_{31t}	0	500	500	500	500
U_{32t}	0	0	500	500	500
U_{33t}	0	0	0	500	500
U_{34t}	0	0	0	0	500

Table B.4 Lower Bound on the Production Level of Potential Generation Units (mw)

Data	**Year1**($t=1$)	**Year2**($t=2$)	**Year3**($t=3$)	**Year4**($t=4$)	**Year5**($t=5$)
L_{5t}	450	450	450	450	450
L_{6t}	0	450	450	450	450
L_{7t}	0	0	450	450	450
L_{8t}	0	0	0	450	450
L_{9t}	0	0	0	0	450
L_{10t}	350	350	350	350	350
L_{11t}	0	350	350	350	350
L_{12t}	0	0	350	350	350
L_{13t}	0	0	0	350	350
L_{14t}	0	0	0	0	350
L_{15t}	300	300	300	300	300
L_{16t}	0	300	300	300	300
L_{17t}	0	0	300	300	300
L_{18t}	0	0	0	300	300
L_{19t}	0	0	0	0	300
L_{20t}	100	100	100	100	100
L_{21t}	0	100	100	100	100
L_{22t}	0	0	100	100	100
L_{23t}	0	0	0	100	100
L_{24t}	0	0	0	0	100
L_{25t}	200	200	200	200	200
L_{26t}	0	200	200	200	200
L_{27t}	0	0	200	200	200
L_{28t}	0	0	0	200	200
L_{29t}	0	0	0	0	200
L_{30t}	200	200	200	200	200
L_{31t}	0	200	200	200	200
L_{32t}	0	0	200	200	200
L_{33t}	0	0	0	200	200
L_{34t}	0	0	0	0	200

Table B.5 Availability Factor of Potential Generation Units

Data	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
a_{5t}	0.95	0.95	0.95	0.95	0.95
a_{6t}	0	0.95	0.95	0.95	0.95
a_{7t}	0	0	0.95	0.95	0.95
a_{8t}	0	0	0	0.95	0.95
a_{9t}	0	0	0	0	0.95
a_{10t}	0.95	0.95	0.95	0.95	0.95
a_{11t}	0	0.95	0.95	0.95	0.95
a_{12t}	0	0	0.95	0.95	0.95
a_{13t}	0	0	0	0.95	0.95
a_{14t}	0	0	0	0	0.95
a_{15t}	0.95	0.95	0.95	0.95	0.95
a_{16t}	0	0.95	0.95	0.95	0.95
a_{17t}	0	0	0.95	0.95	0.95
a_{18t}	0	0	0	0.95	0.95
a_{19t}	0	0	0	0	0.95
a_{20t}	0.95	0.95	0.95	0.95	0.95
a_{21t}	0	0.95	0.95	0.95	0.95
a_{22t}	0	0	0.95	0.95	0.95
a_{23t}	0	0	0	0.95	0.95
a_{24t}	0	0	0	0	0.95
a_{25t}	0.95	0.95	0.95	0.95	0.95
a_{26t}	0	0.95	0.95	0.95	0.95
a_{27t}	0	0	0.95	0.95	0.95
a_{28t}	0	0	0	0.95	0.95
a_{29t}	0	0	0	0	0.95
a_{30t}	0.95	0.95	0.95	0.95	0.95
a_{31t}	0	0.95	0.95	0.95	0.95
a_{32t}	0	0	0.95	0.95	0.95
a_{33t}	0	0	0	0.95	0.95
a_{34t}	0	0	0	0	0.95

Table B.6 Production cost of Potential Generation Units ($/mwh)

Data	**Year1**($t=1$)	**Year2**($t=2$)	**Year3**($t=3$)	**Year4**($t=4$)	**Year5**($t=5$)
F_{5t}	86	87	88	89	90
F_{6t}	0	87	88	89	90
F_{7t}	0	0	88	89	90
F_{8t}	0	0	0	89	90
F_{9t}	0	0	0	0	90
F_{10t}	90	92	94	96	98
F_{11t}	0	92	94	96	98
F_{12t}	0	0	94	96	98
F_{13t}	0	0	0	96	98
F_{14t}	0	0	0	0	98
F_{15t}	92	94	96	98	100
F_{16t}	0	94	96	98	100
F_{17t}	0	0	96	98	100
F_{18t}	0	0	0	98	100
F_{19t}	0	0	0	0	100
F_{20t}	93	94	95	96	97
F_{21t}	0	94	95	96	97
F_{22t}	0	0	95	96	97
F_{23t}	0	0	0	96	97
F_{24t}	0	0	0	0	97
F_{25t}	94	96	98	100	102
F_{26t}	0	96	98	100	102
F_{27t}	0	0	98	100	102
F_{28t}	0	0	0	100	102
F_{29t}	0	0	0	0	102
F_{30t}	95	97	99	101	103
F_{31t}	0	97	99	101	103
F_{32t}	0	0	99	101	103
F_{33t}	0	0	0	101	103
F_{34t}	0	0	0	0	103

Table B.7 Construction Cost of Potential Generation Units (US$10000)

Data	**Year1**($t=1$)	**Year2**($t=2$)	**Year3**($t=3$)	**Year4**($t=4$)	**Year5**($t=5$)
E_{5t}	10000	0	0	0	0
E_{6t}	0	11000	0	0	0
E_{7t}	0	0	12000	0	0
E_{8t}	0	0	0	13000	0
E_{9t}	0	0	0	0	14000
E_{10t}	15000	0	0	0	0
E_{11t}	0	16000	0	0	0
E_{12t}	0	0	17000	0	0
E_{13t}	0	0	0	18000	0
E_{14t}	0	0	0	0	19000
E_{15t}	10000	0	0	0	0
E_{16t}	0	11000	0	0	0
E_{17t}	0	0	12000	0	0
E_{18t}	0	0	0	13000	0
E_{19t}	0	0	0	0	14000
E_{20t}	8000	0	0	0	0
E_{21t}	0	9000	0	0	0
E_{22t}	0	0	10000	0	0
E_{23t}	0	0	0	11000	0
E_{24t}	0	0	0	0	12000
E_{25t}	6000	0	0	0	0
E_{26t}	0	7000	0	0	0
E_{27t}	0	0	8000	0	0
E_{28t}	0	0	0	9000	0
E_{29t}	0	0	0	0	10000
E_{30t}	6000	0	0	0	0
E_{31t}	0	7000	0	0	0
E_{32t}	0	0	8000	0	0
E_{33t}	0	0	0	9000	0
E_{34t}	0	0	0	0	10000

Table B.8 Global Optimal Solution and Integer Variables

Items	**Value**
Objective Value	$25254370000
y_1	1
y_2	1
y_3	1
y_4	1
y_6	1
y_{11}	1
y_{16}	1
y_{20}	1
y_{26}	1
y_{31}	1
Other Integer Variable y_j's	0

Table B.9 Demand-side Management Program(mw)

Variable	**Year1**($t=1$)	**Year2**($t=2$)	**Year3**($t=3$)	**Year4**($t=4$)	**Year5**($t=5$)
u_t	1000	550	1050	1350	754.5

Table B.10 Unused Fund (US$10000)

Variable	**Year1**($t=1$)	**Year2**($t=2$)	**Year3**($t=3$)	**Year4**($t=4$)	**Year5**($t=5$)
S_t	45000	87600	127608	164816.6	202298.3

Table B.11 Production Level of Existing Generation Units (mw)

Variable	**Year1**($t=1$)	**Year2**($t=2$)	**Year3**($t=3$)	**Year4**($t=4$)	**Year5**($t=5$)
x_{1t}	500	350	350	550	550
x_{2t}	200	300	300	300	603
x_{3t}	300	300	300	300	637.5
x_{4t}	400	400	400	400	450

Table B.12 Production Level of Potential Generation Units (mw)

Variable	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
x_{5t}	0	0	0	0	0
x_{6t}	0	450	450	450	855
x_{7t}	0	0	0	0	0
x_{8t}	0	0	0	0	0
x_{9t}	0	0	0	0	0
x_{10t}	0	0	0	0	0
x_{11t}	0	350	350	350	350
x_{12t}	0	0	0	0	0
x_{13t}	0	0	0	0	0
x_{14t}	0	0	0	0	0
x_{15t}	0	0	0	0	0
x_{16t}	0	300	300	300	300
x_{17t}	0	0	0	0	0
x_{18t}	0	0	0	0	0
x_{19t}	0	0	0	0	0
x_{20t}	100	100	100	100	100
x_{21t}	0	0	0	0	0
x_{22t}	0	0	0	0	0
x_{23t}	0	0	0	0	0
x_{24t}	0	0	0	0	0
x_{25t}	0	0	0	0	0
x_{26t}	0	200	200	200	200
x_{27t}	0	0	0	0	0
x_{28t}	0	0	0	0	0
x_{29t}	0	0	0	0	0
x_{30t}	0	0	0	0	0
x_{31t}	0	200	200	200	200
x_{32t}	0	0	0	0	0
x_{33t}	0	0	0	0	0
x_{34t}	0	0	0	0	0

Table B.13 Power Generated and Delivered by Existing Generation Units (mw)

Variable	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
b_{1t}	500	350	350	550	550
b_{2t}	200	300	300	300	603
b_{3t}	300	300	300	300	637.5
b_{4t}	400	400	400	400	450

Table B.14 Power Generated and Delivered by Potential Generation Units (mw)

Variable	Year1($t=1$)	Year2($t=2$)	Year3($t=3$)	Year4($t=4$)	Year5($t=5$)
b_{5t}	0	0	0	0	0
b_{6t}	0	450	450	450	855
b_{7t}	0	0	0	0	0
b_{8t}	0	0	0	0	0
b_{9t}	0	0	0	0	0
b_{10t}	0	0	0	0	0
b_{11t}	0	350	350	350	350
b_{12t}	0	0	0	0	0
b_{13t}	0	0	0	0	0
b_{14t}	0	0	0	0	0
b_{15t}	0	0	0	0	0
b_{16t}	0	300	300	300	300
b_{17t}	0	0	0	0	0
b_{18t}	0	0	0	0	0
b_{19t}	0	0	0	0	0
b_{20t}	100	100	100	100	100
b_{21t}	0	0	0	0	0
b_{22t}	0	0	0	0	0
b_{23t}	0	0	0	0	0
b_{24t}	0	0	0	0	0
b_{25t}	0	0	0	0	0
b_{26t}	0	200	200	200	200
b_{27t}	0	0	0	0	0
b_{28t}	0	0	0	0	0
b_{29t}	0	0	0	0	0
b_{30t}	0	0	0	0	0
b_{31t}	0	200	200	200	200
b_{32t}	0	0	0	0	0
b_{33t}	0	0	0	0	0
b_{34t}	0	0	0	0	0

Table B.15 Power Generated but Not Delivered by Existing Generation Units (mw)

Variable	**Year1**($t = 1$)	**Year2**($t = 2$)	**Year3**($t = 3$)	**Year4**($t = 4$)	**Year5**($t = 5$)
d_{1t}	0	0	0	0	0
d_{2t}	0	0	0	0	0
d_{3t}	0	0	0	0	0
d_{4t}	0	0	0	0	0

Table B.16 Power Generated but Not Delivered by Potential Generation Units (mw)

Variable	**Year1**($t=1$)	**Year2**($t=2$)	**Year3**($t=3$)	**Year4**($t=4$)	**Year5**($t=5$)
d_{5t}	0	0	0	0	0
d_{6t}	0	0	0	0	0
d_{7t}	0	0	0	0	0
d_{8t}	0	0	0	0	0
d_{9t}	0	0	0	0	0
d_{10t}	0	0	0	0	0
d_{11t}	0	0	0	0	0
d_{12t}	0	0	0	0	0
d_{13t}	0	0	0	0	0
d_{14t}	0	0	0	0	0
d_{15t}	0	0	0	0	0
d_{16t}	0	0	0	0	0
d_{17t}	0	0	0	0	0
d_{18t}	0	0	0	0	0
d_{19t}	0	0	0	0	0
d_{20t}	0	0	0	0	0
d_{21t}	0	0	0	0	0
d_{22t}	0	0	0	0	0
d_{23t}	0	0	0	0	0
d_{24t}	0	0	0	0	0
d_{25t}	0	0	0	0	0
d_{26t}	0	0	0	0	0
d_{27t}	0	0	0	0	0
d_{28t}	0	0	0	0	0
d_{29t}	0	0	0	0	0
d_{30t}	0	0	0	0	0
d_{31t}	0	0	0	0	0
d_{32t}	0	0	0	0	0
d_{33t}	0	0	0	0	0
d_{34t}	0	0	0	0	0

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